SAMPLE EXAM QUESTIONS
ENGINEERING PROJECT MANAGEMENT & FINANCE
(UESTC 3031& UESTCHN3012)

SAMPLE SET 1

Section A: Multiple Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options for each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) Which item below is correct regarding Design for Manufacturing (DFM): [2]

A. As a general rule, DFM can't be started at the initial stages of product design before fabrication.

B. DFM has advantages, including lower production costs and higher product quality; however, it may increase the time required to bring the product to market.

C. The DFM may impose higher manufacturing costs, but instead, it can benefit from higher customer satisfaction because of improved product quality.

D. DFM is a design method to reduce the complexity of manufacturing operations and the overall cost of production including the cost of raw materials.

E. While manufacturers were facing strong competition from other manufacturers (rivals), at the time, they were under pressure to increase production efficiency while increasing costs of production.

(1.2) There is a variety of actionable sustainable design principles that can be implemented to create better designs. Some of them are below except: [2]

A. Dematerialization.

B. Migration to product-service systems.

C. Limit or eliminate long-distance outsourcing.

D. Design for a shorter period of usage.

E. Invest in simulation.

(1.3) Which of the following 7QC tools can best be used to logically organize possible factors related to a specific problem by graphically displaying them in increasing detail and suggesting causal relationships among theories? [2]

A. Cause-Effect Diagram.

B. Check Sheet.

C. Pareto Chart.

D. Control Charts.

E. Scatter Diagram.

(1.4) Which model below is best used as a roadmap for Six Sigma to improve the quality of results produced by company processes? [2]

A. Define, Measure, Analyze, Improve and Control (DMAIC)

B. Plan, Do, Check, Act (PDCA)

C. Total Quality Management (TQM)

D. Lean Manufacturing

E. Failure Modes and Effects Analysis (FMEA)

(1.5) Project A has an NPV of $1.2 million; Project B has a BCR of 1.035; Project C has an IRR of -0.03 percent. Based on the information given, which project appears to be most attractive? [2]

A. Project A is the most attractive.

B. Project B is the most attractive.

C. Project C is the most attractive.

D. All three projects A, B, and C are equally attractive to execute.

E. It is uncertain which project is the most attractive to execute based on the provided information.

(1.6) You are a project manager for an electrical manufacturing company involved in developing innovative components for renewable energy solutions. You have noticed that one of your team members is lacking knowledge in electronic component design. The same was communicated to him, and you tried sending him for training. However, the team member started arguing that he is an expert in electronic component design. Which of the following best describes this situation? [2]

A. The team member is consciously competent.

B. The team member is unconsciously competent.

C. The team member is consciously incompetent.

D. The team member is unconsciously incompetent.

E. The team member is aware of his/her skill, so there is no necessity for training.

(1.7) In a Profit and Loss (Income) statement, Earnings before Interest, Tax, Depreciation, and Amortization (EBITDA) is equal to: [2]

A – Total Sales minus Cost of Sales

B – Total Expenses minus Net Profit

C – Total Sales minus (Salaries + Rent)

D – Gross Profit minus all company operating expenses (salaries and rent etc.)

E – Full production costs minus Net Profit

(1.8) You are an engineer working on a project that depends on a new type of battery technology for success. During the development of the product, a report appears in a respected science publication that there may be health issues associated with the new battery technology if it leaks but they do not yet have reliable data. You are under pressure from your boss to complete your project, what do you do? [2]

A – Tell your boss there is no reliable data so you should complete the project against the original plan. It is not your fault if there is no reliable data.

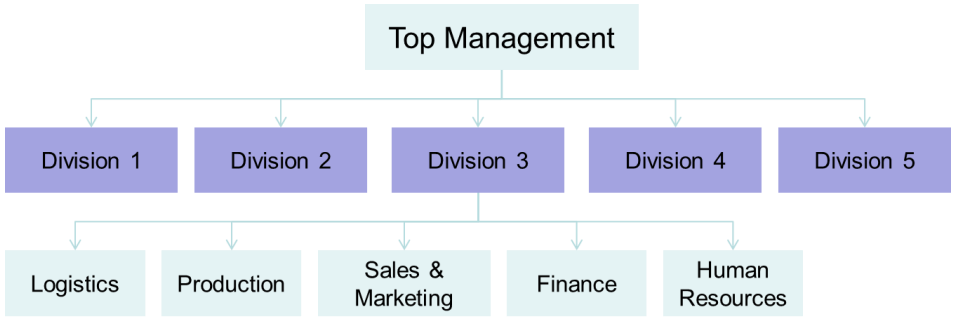
B – Advise your boss that although there is no reliable data, you should adopt a precautionary approach and install further protection (costing more money) around the battery along with a monitoring system to detect leaks.

C – Stop work on the project and advise your boss that if he does not cancel the project you will report him to the health authorities.

D – Advise your boss if you cannot use the battery technology then they should cancel the project and the product that uses it as it will not be competitive in the market.

E – Approach the author of the science publication and offer to pay for a holiday for them and their family if the author writes another article stating that the first article was wrong.

(1.9) If a company has an organizational structure shown below, it is called: [2]



A – A Team-Based Structure

B – A Matrix Structure

C – A Divisional Structure

D – A Simple Startup Structure

E – A Military Command Structure

(1.10) You have fifteen members in your engineering project team. You have called for a project status meeting without prior notice. During the meeting, there were too many side conversations and arguments. As a result, the team achieved very little in the meeting. To achieve orderliness in the meeting next time, you should: [2]

A – Decrease the number of people in the meeting

B – Publish a meeting agenda

C – Ensure that you control the channels of communication

D – Give incentives to team members for conforming to desired meeting norms

E – Invite a member from the higher management/executive team

(1.11) A part of the cash flow with 30-days net payment terms for cash-in and cash-out is given in Table Q1 (1.11) below. Determine the cumulative cash towards the end of each month (M). The currency utilized for transactions in this cash flow is RMB. [2]

Table Q1 (1.11): Cash-Flow Table

	M0	**M1**	**M2**	**M3**	**M4**	**M5**
Sales Bookings	10	12	8	20	100	0
Shipments	0	10	12	8	20	100
Components Order	5	6	4	10	50	0

A – M0 = 0 RMB; M1 = -5 RMB; M2 = -1 RMB; M3 = +7 RMB; M4 = +5 RMB; M5 = -25 RMB

B – M0 = -5 RMB; M1 = -6 RMB; M2 = -4 RMB; M3 = -10 RMB; M4 = -50 RMB; M5 = 0 RMB

C – M0 = 0 RMB; M1 = -10 RMB; M2 = -12 RMB; M3 = -8 RMB; M4 = -20 RMB; M5 = -100 RMB

D – M0 = -5 RMB; M1 = -16 RMB; M2 = -16 RMB; M3 = -18 RMB; M4 = -70 RMB; M5 = -100 RMB

E – M0 = 0 RMB; M1 = -5 RMB; M2 = +4 RMB; M3 = +8 RMB; M4 = -2 RMB; M5 = -30 RMB

(1.12) Based on Q1 (1.11) above, how do you classify the trading nature for such a type of cash-flow statements? [2]

A – Well-balanced trading

B – Under trading

C – Over trading

D – No trading

E – Well-established trading

Section B: Long Questions

Q2 [25 marks]

(a) A company is designing a new household appliance and is considering two possible designs for one of its components.

Design A consists of five separate parts that need to be assembled.

Design B integrates these parts into a single molded component.

(i) Discuss the advantages and disadvantages of each design from a Design for Manufacturing (DFM) perspective. [6]

(ii) Design A consists of five separate parts that need to be assembled, and one of these parts can be reused in other devices. Design B integrates all parts into a single molded component, making it smaller in size. In terms of actionable sustainable design principles, compare the two designs. [4]

(iii) The 4Rs, in particular, are a very effective tool in building a circular economy. They include four principles that can be applied by almost every individual and have an exponential effect. Distinguish the principles of the 4Rs that Design A and Design B primarily apply to promote the circular economy. [3]

(b) The night shift operator recorded twenty current (mA) samples during the electronic circuit testing over time in electronics manufacturing, as shown in Table Q2(b). As a technician, you are expected to evaluate and analyze the process over the last 20 time units to determine whether the process is under control (stable) or not. Discuss your findings by utilizing a control chart (considering 2 * Sigma). You are required to create a plot to demonstrate your explanation with a visual representation that includes the central line, upper control limit (UCL), and lower control limit (LCL). [7]

#Sample	Current (mA)	#Sample	Current (mA)
1	510	11	505
2	516	12	511
3	500	13	500
4	495	14	498
5	505	15	501
6	500	16	500
7	510	17	490
8	505	18	505
9	500	19	507
10	499	20	500

Table Q2(b)

(c) Assume that the manufacturer's target for the process in (b) is 500 mA, with a tolerance of ±5 mA. To assess the manufacturing process's capability to meet specifications, evaluate both process capability indices for this manufacturing operation. Discuss your findings. [5]

Q3 [25 marks]

(a) Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements. Answer the following questions.

(i) John is overseeing the development of a new electronic device prototype. After collaborating with the team, they determine the most likely project duration of 10 months. In the best-case scenario, efficient collaboration with a specialized component supplier could reduce the duration by 2 months. However, a delay in obtaining critical components might extend the project by 3 months. Using the three-point estimating technique, calculate the expected project duration using both Triangular Distribution and Beta Distribution. Also, determine the standard deviation for each of these approaches. [3]

(ii) You are leading a project team consisting of 11 members. As the project transitions into the next phase, three new members join the team while two existing members depart. Based on these changes, determine the number of additional communication channels that will be created. [2]

(iii) Assume an electronic product engineer has collected 10 observations (test results) for a specific product's quality characteristic, which is nominally the best at a target value of 8.5. The engineer uses the Taguchi loss function to evaluate the quality, where the constant k=1. Given the data provided in Table Q3(a), compute the Taguchi loss function for each observation and determine the total loss. [3]

Observation No.	Measurement (Quality Characteristic)	Observation No.	Measurement (Quality Characteristic)
1	8.1	6	8.35
2	8.9	7	8.25
3	8.45	8	8.68
4	9.25	9	8.9
5	8.86	10	9.05

Table Q3(a)

(iv) You have been assigned as the project manager for the development of a basic embedded system for controlling a temperature sensor in a smart home application. The major phases of the project are planning, design, and testing. The WBS is decomposed down through these phases, where each phase is broken down into two additional activities at Level 2. Based on this, plot the WBS for the project, showing all three levels clearly. [4]

(b) Project schedule management includes the processes required to manage the timely completion of the project. One of the common techniques used is CPM (critical path method). The critical path is the sequence of activities that represents the longest path through a project, which determines the minimum possible project duration. Consider a project (assume date 0 for project starting) with the network diagram shown in Figure Q3(b) and answer the following questions.

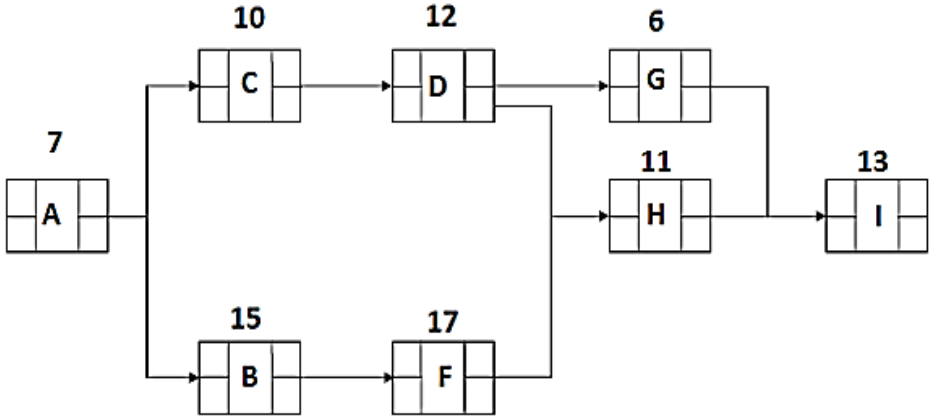


Figure Q3(b)

(i) Perform the forward pass analysis through the schedule network and compute Early Start and Early Finish dates for all activities. [5]

(ii) Perform the backward pass analysis through the schedule network and compute Late Start and Late Finish dates for all activities. [5]

(iii) Calculate the total float (slack) for all activities. [2]

(iv) Define the Critical Path (sequence of activities that are located on the critical path). [1]

Q4 [26 marks]

VRtuoso is a company that produces virtual reality (VR) headsets for businesses and academic (teaching and research) purposes wherein each headset comprises of two specialized lenses. The company manufactures 0.6 Million (M) headsets annually. The lenses for these headsets are provided by an external supplier to VRtuoso for 675 RMB each. Moreover, the company has some vacant production facilities that adds to the total annual costs with a Fixed Overhead Cost of 39 M RMB. Currently, the vacant production facilities are rented out for 2 M RMB per year but the company has recently been informed during a TV interview that they might not renew the rental agreement for the next term.

(a) Draw a table comparing the equivalent cost of purchasing a lens from the external supplier while the vacant production facilities are rented out vs. the scenario if the rental agreement for the vacant production facilities is not renewed for the next term. [5]

(b) VRtuoso drops the plan of renewing the rental agreement for the vacant production facilities instead plans to utilize the vacant production facilities for manufacturing the lenses within the company. In addition to the Fixed Overhead Cost, the production cost breakdown for a lens consists of a direct materials cost of 325.00 RMB, a direct labor cost of 215.00 RMB, and a variable overhead cost of 127.50 RMB. Given this new information, prepare a new cost table demonstrating the annual impact of

manufacturing the lenses within the production facilities available in the company. Should VRtuoso continue purchasing the lenses from the external supplier or start manufacturing their own lenses? [5]

(c) If VRtuoso can sell the VR headsets for 1,413 RMB each, based on the information given above in Q4(b), draw a table demonstrating the production costs and sales revenue of manufacturing and selling VR headsets from $0 \rightarrow 1$ M in quantity with an increment size of 0.2 M VR headsets. [6]

(d) If the VRtuoso factory is running at full capacity of 0.6 M VR headsets per year with an assumed price of 1,413 RMB per VR headset, what is the profit made by VRtuoso? What is the safety margin for this scenario? [5]

(e) Draw a Break-even graph showing Total Cost (TC), Variable cost (VC), and Revenue (REV) against production quantity (Q). Calculate the Break-even Point and indicate this on the graph. [5]

SAMPLE SET 2

Section A: Multiple Choice Questions [24 marks]

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options for each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) Which of the following best aligns with the principles of Design for Manufacturing (DFM)? [2]

A. Increasing the number of parts in a design to improve flexibility in manufacturing.

B. Using complex shapes and intricate structures to enhance product uniqueness.

C. Minimizing the number of parts and using straightforward shapes to ease production and reduce costs.

D. Prioritizing visual design over how easy it is to make.

E. Adding redundant features to ensure product robustness.

(1.2) Which statement accurately describes a cause-effect diagram? [2]

A. A cause-effect diagram is primarily used to illustrate the effects of a specific problem without identifying its root causes.

B. Cause-effect diagrams, also known as fishbone or Ishikawa diagrams, are visual tools used to focus on the sole cause of a given problem.

C. A cause-effect diagram called the fishbone or Ishikawa diagram is designed to indicate the relationships between various effects easily with no need for visual tools.

D. A cause-effect diagram is a visual tool used to logically organize possible causes for a specific problem or effect by graphically displaying them in increasing detail, suggesting causal relationships among theories.

E. Cause-effect diagrams called fishbone or Ishikawa diagrams are visual tools exclusively used to identify effects, without the need for organizing possible causes or establishing causal relationships.

(1.3) Which of the following 7QC tools can best be used to study how a process changes over time? [2]

F. Cause-Effect Diagram.

G. Check Sheet.

H. Pareto Chart.

I. Control Charts.

J. Scatter Diagram

(1.4) Which of the following is not considered a project? [2]

F. Creating a prototype for a next-generation smartphone.

G. Designing and deploying a new software algorithm for real-time data processing in an electronic control system

H. Developing periodic updates for an existing software system in a manufacturing plant.

I. Setting up a waste reduction program in an office environment.

J. Designing and constructing a printed circuit board (PCB) for a new electronic device.

(1.5) This technique can be mainly used in project management to identify risks, particularly in assessing external factors and internal vulnerabilities that can impact the project. [2]

A. Failure Modes and Effects Analysis (FMEA)

B. Strengths, Weaknesses, Opportunities, and Threats (SWOT)

C. Internal Rate of Return (IRR)

D. Net Present Value (NPV)

E. Benefit-Cost Ratio (BCR)

(1.6) Carmen is just about to deliver the case study commissioned by the Tesla Gigafactory project. Before she does, she arranges a meeting with the steering committee and project sponsor associated with the Tesla Gigafactory to take them through the case study, answer any questions or concerns they may have, and seek to get official confirmation that the case study meets their expectations. What process is she performing here? [2]

A. Deliverables Acceptance.

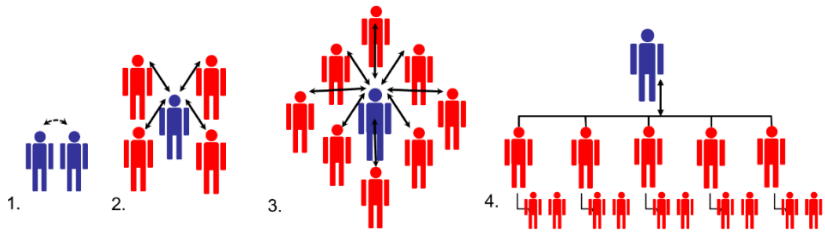
B. Accept Deliverables.

C. Validate Scope.

D. Plan Scope Management.

E. Close Project or Phase.

(1.7) The following organizational structures should be labeled as: [2]



A – 1. Start-up; 2. Expansion; 3. Growth; 4. Formal Organisation

B – 1. Expansion; 2. Start-up; 3. Formal Organisation; 4. Growth

C – 1. Start-up; 2. Growth; 3. Expansion; 4. Formal Organisation

D – 1. Expansion; 2. Start-up; 3. Growth; 4. Formal Organisation

E – 1. Growth; 2. Start-up; 3. Formal Organisation; 4. Expansion

(1.8) Professional engineers have a significant role to play in ensuring that the projects they contribute to have been developed with sustainability in mind and should apply the 6 principles for sustainability to all their projects. One of those principles is to 'Use resources efficiently and effectively'. Which of the following statements does NOT adhere to this principle? [2]

A – Understand there are environmental limits and finite resources

B – Use systems and products that reduce embedded carbon, energy and water use, waste, and pollution

C – Engage a waste management contractor who specializes in shipping waste to regions with lower environmental standards

D – Adopt strategies for re-use, recycling, decommissioning, and disposal of components and materials

E – Work to repair any damage

(1.9) Consider the structure of a start-up company run by an entrepreneur. Find the correct pair of true statements [2]

1 Maximum standardisation and formalisation
2 Few layers: Limited middle-line managerial levels
3 Decentralized and indirect supervision
4 Wide span of control (around the entrepreneur)

A – 1 and 2

B – 3 and 4

C – 2 and 3

D – 2 and 4

E – 1 and 3

(1.10) It costs 1000 RMB for hand tools and 1.50 RMB labor per unit to manufacture a product. An alternative is to manufacture the product with automated machinery costing 15,000 RMB, with a 0.50 RMB per-unit cost. With an annual production rate of 5000 units, how long will it take to reach the production cost cross-over point? [2]

A – 2.0 yr

B – Never

C – 3.6 yr

D – 2.8 yr

E – 0.9 yr

(1.11) When developing a new product, which stage has the biggest impact on the final product's price, quality, and manufacturing time? [2]

A – Mostly during manufacturing

B – Mostly during product development

C – Equally during product development and manufacturing

D – Mostly during sales and marketing

E – After customer feedback during mass production

(1.12) In the execution phase of a project, key man resources are lost, and new staff join. The project manager sets up a meeting with the new team. What should be the most important agenda item for this meeting? [2]

A – Discussing team building activities

B – Reviewing WBS and responsibilities of all team members

C – Inviting the team members to share their past project experiences

D – Planning risk responses for existing risks

E – Assigning someone from the new members to lead the meeting

Section B: Long Questions

Q2 [26 marks]

(d) Sustainability consists of fulfilling the needs of current generations without compromising the needs of future generations while ensuring a balance between the three sustainability pillars.

(i) Explain these three pillars of sustainability. [6]

(ii) Compare the Responsibility for Sustainability between Manufacturing Companies and End-User Consumers. [4]

(e) In a circuit board assembly line, the production team has noticed an increasing number of incorrect signal outputs on the assembled boards. These failures occur during the signal testing phase, where the boards are not performing according to their specifications. The issue has been identified recently, and the failure rate seems to be higher in some batches. Your task is to investigate the root causes of this issue using a Cause and Effect (Fishbone) Diagram. Plot the Fishbone Diagram, focusing on the most relevant categories: Machines, Materials, Methods, and Manpower. List two potential causes under each category that could be contributing to the incorrect signal output issue. [6]

(f) Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.

(i) In project management, what is the purpose of the Benefit-Cost Ratio (BCR)? Given that Project A has a BCR of 1, what can we infer about this project? [3]

(ii) What is the term used to describe the process of developing a detailed description of the project and product, often represented as a hierarchical decomposition of the project scope? [1]

(iii) As a project manager of an EPC company, you have identified all risks initially and arranged them according to the category. What should you do next? [1]

(g) In conflict management, understanding various strategies is crucial, such as Competing, Collaborating, Avoiding, Accommodating, and Compromising. Each strategy leads to different win or lose outcomes for the parties involved in the conflict. Based on the scenarios below, identify the conflict resolution strategy used and determine whether each side of the conflict experiences a win or a loss.

(i) Two team members are arguing over which software tool to use for tracking project progress. The project manager sees that the argument is taking too much time and decides to postpone the discussion indefinitely, hoping the issue will resolve itself. [1]

(ii) A senior stakeholder dislikes a minor feature in the project, but the project manager knows that arguing over it could damage the relationship. To maintain harmony, the manager agreed to remove the feature, even though it was beneficial. [1]

(iii) The marketing and development teams disagree on the launch date. The developers need more time for testing, while marketing wants to meet a deadline for a big campaign. Instead of arguing, both teams work together to adjust the scope, ensuring essential features are tested while still hitting the marketing deadline. [1]

(iv) A critical bug is delaying the project. The lead engineer insists on a lengthy, detailed fix, while the project manager, under pressure from upper management, orders an immediate workaround to meet the deadline. [1]

(v) Two departments are fighting over resource allocation. One wants five additional developers, while the other needs three. The project manager negotiates, giving each team some of what they want, one team gets three developers, and the other two. [1]

Q3 [25 marks]

As a young entrepreneur, you would like to guarantee a healthy total cost of ownership by evaluating the net present value of your young company while utilizing discounted cash flow analysis.

(f) Consider your company is designing a new product that will take your company three years to design, fabricate, and build. For this product, your company's expenses are 1,000,000 RMB per year, which will increase with an inflation rate of 5% annually. How much must your company quote your customer (for delivery in three years' time) if your company would like to cover their expenses and earn a net profit (in today's equivalent currency) of 100,000 RMB? [3]

(g) Consider your company would like to purchase a new machine that has a purchase price of 10 Million (M) RMB with a lifetime of five years. The machine will require annual maintenance costs of 1 M RMB associated with annual power costs of 0.5 M RMB. All costs are fixed with inflation of 10% applied to power only after year 1.

i) Demonstrate the total projected costs over the lifetime of this new machine. [3]

ii) Analyze the total cost of ownership over the lifetime of this new machine. [3]

(h) Consider you have decided that your company will make a significant investment into expanding its presence in China by setting up a large assembly facility in Chengdu. Your company has estimated its initial set-up costs to be in the region of 6,398 M RMB. Net income from this new assembly facility is forecasted as follows:

Year (Y) 1: 1,400 M RMB
Y2: 1,450 M RMB
Y3: 1,550 M RMB
Y4: 1,625 M RMB
Y5: 1,480 M RMB

i) Calculate the projected payback time for this new assembly facility to the nearest month. [5]

ii) Calculate the net present value of this new assembly facility based on a discounting factor of 5%. [5]

iii) Comment on the attractiveness of this new assembly facility. [3]

(i) Consider a part of cash-flow with 30 days net payment terms for cash-in and cash-out given in Table Q2 below. Determine the cumulative cash towards the end of each month (M). The currency utilized for transactions in this cash flow is RMB. [3]

Table Q2: Cash-Flow Table

	M0	**M1**	**M2**	**M3**	**M4**	**M5**
Sales Bookings	10	12	8	20	100	0
Shipments	0	10	12	8	20	100
Components Order	5	6	4	10	50	0

Q4 [25 marks]

You work for a Chinese telecom company that supplies the industry with Wi-Fi routers. The company has an annual production capacity of 45,000 routers. Your production cost breakdown for 45,000 routers is shown in Table Q4 below.

Table Q4: Cost breakdown for 45,000 routers

Cost category	**Annual Cost (RMB)**
Direct Materials	405,000
Direct Labour	225,000
Variable Overhead	45,000
Fixed Overhead Cost	562,500
Total	**1,237,500**

An outside supplier has approached your company and offered to supply the Wi-Fi routers to your company for 27.5 RMB each. Your manager has asked you to carry out an analysis of this offer. He advises you that the Fixed Overhead Costs for the production facilities will NOT be avoided if you purchase the routers from this outside supplier as your company has no other use for the vacant production facilities if you purchase from the outside supplier.

(a) Draw a table comparing the cost of making the routers internally on a per router basis with the costs of purchasing the routers from the outside supplier. Do you recommend that your company purchases from an outside supplier? [5]

(b) Your manager now tells you he can rent out the production facilities for 570,000 RMB per year. Given this new information, prepare a new cost table showing the impact of the facilities rental. Does this information change your recommendation on whether to purchase from an outside supplier? [5]

(c) Using the information provided in Table Q4 and knowing that:

TC = FC + (Q * VC),

where

TC = Total Cost
FC = Fixed Cost
VC = Variable Cost = (Direct Material + Direct Labour + Variable Overhead) all per router
Q = Quantity Produced

If your company can sell the Wi-Fi routers for 40.00 RMB each, draw a table showing the production costs and sales revenue of manufacturing and selling router quantities from $10{,}000 \rightarrow 30{,}000$ routers in 5000 router increments. [5]

(d) If the factory is running at full capacity of 45,000 routers per year with an assumed price of 40.00 RMB per router, what is the profit made by the company? Also, calculate the safety margin for this scenario. [5]

(e) Draw a Break-even graph showing Total Cost (TC), Variable cost (VC), Fixed Cost (FC), and Revenue (REV) against production quantity (Q). Calculate the Break-even Point and indicate this on the graph. [5]

SAMPLE SET 3

Section A: Multiple Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options to each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) Why do companies use project management? Choose the correct combination of statements. [2]

1 - To define the project and agree with the customer

2 - To estimate project cost and make proposals

3 - To assess risk and failure points and make back up plans

4 - To allocate the right resource at the right time

5 – To track the schedule only once the plan is made

A – 1 + 2 + 3 + 5

B – 1 + 2 + 4 + 5

C – 2 + 3 + 4 + 5

D – 1 + 3 + 4 + 5

E – 1 + 2 + 3 + 4

(1.2) In a Profit and Loss (Income) statement, Earnings before Interest, Tax, Depreciation, and Amortisation (EBITA) is equal to: [2]

A – Total Sales minus Cost of Sales

B – Total Expenses minus Net Profit

C – Total Sales minus (Salaries + Rent)

D – Gross Profit minus all company operating expenses (salaries and rent etc.)

E – Full production costs minus Net Profit

(1.3) If a company has an organisational structure as shown below in Figure Q1(1.3), it is called: [2]

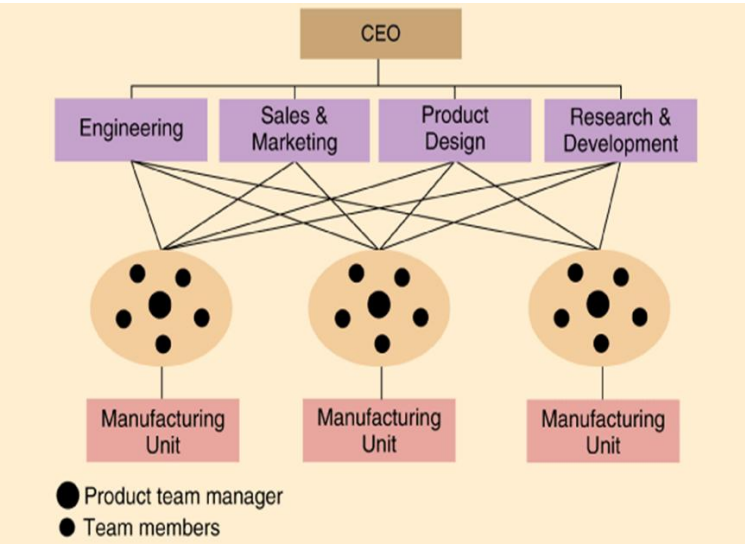


Figure Q1(1.3)

A – A Team-Based Structure

B – A Matrix Structure

C – A Divisional Structure

D – A Simple Startup Structure

E – A Military Command Structure

(1.4) A Sole Trader Company is: [2]

A – A business you run as an individual. You can keep all your business' profits after you have paid tax on them. You are personally responsible for any losses your business makes.

B – A business in which you and your business partner (or partners) personally share responsibility for your business. You can share all your business's profits between the partners. Each partner pays tax on their share of the profits.

C – A company whose shares can be freely traded on the stock market.

D – An organisation you can set up to avoid paying tax or having any responsibility as a director.

E – An organisation you can set up to run your business- it is responsible for everything it does, and its finances are separate from your personal finances. Any profit it makes is owned by the company after it pays Corporation Tax. The company can then share its profits amongst its shareholders.

(1.5) In a simple Lease–or-Buy calculation, you require a radar equipment for a project. The cost to buy the equipment is 30,000 RMB plus an additional 500 RMB per day. You are able to lease the equipment from a local company at a cost of 2000 RMB per day. What is the maximum length of time you can lease the equipment before it is cheaper to buy the equipment? [2]

A – 3 days

B – 10 days

C – 20 days

D – 30 days

E – 16.333 days

(1.6) If **TC** = Total Cost, **F** = Fixed Costs, **VC** = Variable Cost, **SP** = Selling Price, **R** = Revenue, and **Q** = Quantity of products sold, then the breakeven point of a project is defined as: [2]

A – When **F – (VC)*Q = (SP)*Q**

B – When **TC = R**

C – When **(R)*Q = SP – VC**

D – When **TC = F + (VC)*Q**

E – When **R = (SP)*Q**

(1.7) What is a Profit & Loss (Income) statement? [2]

A – It only summarises all the sales revenues for the financial year

B – It only summarises all the payments or expenses for the financial year

C – It is only used when the company makes a loss

D – It summarises the sales revenues, payments, and expenses for the financial year

E – It is only used when the company makes a profit

(1.8) In a Profit and Loss (Income) statement, the Gross Profit is equal to: [2]

A – Total Expenses minus Net Profit

B – Sales minus Expenses

C – Total Sales minus Cost of Sales

D – Total Sales minus (Salaries + Rent)

E – Full production costs minus Net Profit

(1.9) If a company is described as having a 'Defender (i.e. market leader or stable market)' Strategic Posture, it displays the following characteristics: [2]

	Emphasis	**Market Focus**	**Structure and Process**	**Technology**
A -	Pioneering changes and innovation	Aggressively explore and exploit new	Organic: informal, decentralised, and flexible	Contingent, multiple, prototypical technologies;

		market opportunities		low degree of routinisation and mechanisation
B -	Stability and efficiency	Maintain a small niche	Mechanistic: hierarchical, centralised, and control	One core highly cost-efficient technology
C -	Balanced: minimised risks and maximise profits	Defend core business while cautiously exploit proven new markets by imitating Prospectors	Matrix organisation – coalition between core functions	Dual core with emphasis on applied research
D -	Reluctant to act due to instability and uncertainty	Unsure	Maintain whatever structure is in place; often a weak link with strategy, which is unclear	Unsure
E -	Stability and efficiency	Aggressively explore and exploit new market opportunities	Organic: informal, decentralised, and flexible	Dual core with emphasis on applied research

(1.10) Design for Manufacturing (DFM) is an important part of any product development cycle (e.g., initial design, final design, fabrication, production, and product launch). Which item below is correct: [2]

A – The DFM is more effective to be applied in the production stage because of the higher impact and lower cost of changes due to the product being manufactured and can be changed promptly.

B – The DFM is more effective to be applied during product launch because of the highest impact and lowest cost of changes.

C – The DFM is more effective to be applied to the initial design because of the highest impact and lowest cost of changes.

D – The DFM is more effective to be applied after the initial design and before production when the final design is being finalized and can effectively implement required changes according to customer requirements.

E – The DFM is more effective to be applied in all stages after design because we are sure that the customer requirements are recognized completely and any changes in the future do not impose higher costs of changes.

(1.11) There is a variety of actionable sustainable design principles that can be implemented to create better designs. Some of them are below except: [2]

A – Dematerialization.

B – Migration to product-service systems.

C – Limit or eliminate long-distance outsourcing.

D – Design for a shorter period of usage.

E – Invest in simulation.

(1.12) When analyzing broad causes by looking at the specific parts, the Pareto chart is one of the applicable tools for the control of quality. Which of the following statements about a Pareto chart is true? [2]

A – It is a bar graph that displays the relative frequency or size of problems in descending order of importance.

B – It is a line graph that shows the cumulative percentage of problems plotted against the total number of occurrences.

C – It is a scatter plot that indicates the correlation between two variables.

D – It is a pie chart that breaks down the total number of problems into categories.

E – None of the above statements.

Section B: Long Questions

Q2 You are a project manager working on China's high speed railway. You need to install a major new piece of electrical equipment into a new station that is being built at Chengdu's new airport. The only way to install the new equipment is to hire a tower crane to lift it into position. As the project manager, you have collected all the information and tasks required to build a tower crane on the station site and recorded the tasks in Table Q2(a).

Table Q2(a): Tower Crane tasks and durations.

Task Number	Task Description	Duration (days)	Predecessors
0	Project Start (Milestone)	0	
1	Specify Crane	3	0
2	Survey Site	4	1
3	Survey Power Supply	3	1
4	Select Site	1	2
5	Order Tower Crane	2	3,4
6	Arrange Transport	4	5
7	Order Mobile Erector Crane	3	5
8	Install Concrete Plinth	5	5
9	Confirm Site Access	1	6,7,8
10	Get Mobile Erector Crane on Site	4	9
11	Crane Parts on Site	5	9
12	Install Crane Base	2	10,11
13	Install Tower & Crane Head	1	12
14	Assemble Jib	2	10,11
15	Install Jib	1	12,13
16	Install counterweight	1	15
17	Apply Crane Power	1	16
18	Test Crane	1	17
19	Project end	0	18

(a) Show the above information on a Network diagram showing each task duration and the interdependencies of each task. [8]

(b) Using the Network diagram:

(i) List the critical path tasks in the order they occur. [3]

(ii) Calculate the project's overall duration (show your working). [2]

(c) Develop a Gantt chart for this project showing all tasks, interdependencies, slack time, and critical path. [7]

(d) If Task 8 (Install Concrete Plinth) cannot be started until the 'Mobile Erector Crane' has been ordered (Task 7) has been completed:

(i) Calculate the new project duration and describe how this change impacts the critical path. [2]

(ii) Comment on the usefulness of a Gantt chart to a project manager when changes occur. [3]

Q.3

(h) Design For Manufacturing (DFM) consists of some main principles such as Simplicity, Standardization, Tolerance, Material Selection, Automation, and Process Integration.

(iv) Describe briefly any two of these DFM principles. [4]

(v) The 4Rs in particular are very effective tools in building a circular economy for sustainable products. It includes four principles that can be applied by almost every individual and have an exponential effect. List any two of these principles. [2]

(i) Quality control (QC) is a procedure or set of procedures intended to ensure that a manufactured product or performed service adheres to a defined set of quality criteria or meets the requirements of the client or customer.

(i) By using seven basic tools of quality, we can manage the quality of our product or process effectively. Known around the world as the seven-quality control (7-QC) tools. List any four of these tools. [4]

(ii) Let's say you are a manufacturer of printed circuit boards (PCBs) for electronic devices. You want to monitor the defect rate of your PCBs to ensure that they meet the required quality standards. You decide to monitor the defect rate of your PCBs. After collecting data for a period of time, you plot the data and check the trend of the defect rate over time. Which quality tool is used? [1]

(iii) In the schematic quality tool that you used in above (ii), what exactly do you see if noticed that the process is unstable? [2]

(j) Robust Design is an engineering methodology for improving productivity during research and development so that high-quality products can be produced quickly and at a low cost.

(i) The four different versions of the quadratic loss function are plotted in Figure Q3 based on the Taguchi overview. List the name of any two of these loss functions. [2]

(ii) In robust design terminology, the performance of a product as measured by the

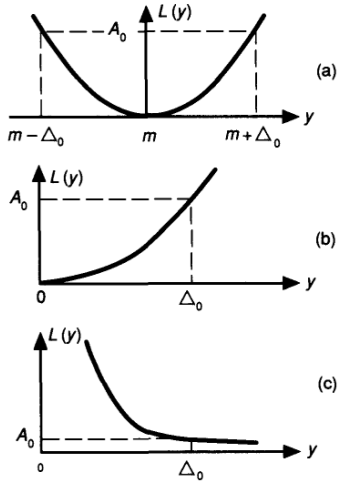


Figure Q3

quality characteristic varies in the field due to a variety of causes. We call all such causes noise factors, and they can be generally classified into three types. Explain any two types of noise factors. [4]

(k) Six-sigma is a statistical concept that measures a process in terms of defects – at the six sigma level, there are 3.4 defects per million opportunities:

(i) List any of the five main phases in 6-Sigma. [2]

(ii) There are five key roles for Six Sigma in manufacturing including Executive Leadership, Champions, Master Black Belts, Black Belts, and Green Belts. Explain briefly any two of these roles. [4]

Q4

(a) Derive the theoretical break-even revenue relationship in terms of fixed costs, variable costs, and selling price. [3]

You work for XYZ, a company that requires 17,500 optical transceivers annually to manufacture its optical wireless communication (OWC) systems. These optical transceivers are supplied by an external supplier to your company for 57 RMB each. Moreover, your company has some vacant production facilities that adds to the total annual costs with Fixed Overhead Cost of 568,750 RMB. Currently, the vacant production facilities are rented out for 225,000 RMB per year but your manager has informed you that the rental agreement is nearing an end soon.

(b) Draw a table comparing the equivalent cost of purchasing a transceiver from the external supplier while the vacant production facilities are rented out with the scenario when the rental agreement has expired. [5]

(c) Your manager now tells you he wants to utilise the vacant production facilities and manufacture the transceivers within the company. In addition to the Fixed Overhead Cost, the production cost breakdown for a transceiver is shown in Table Q4(c) below.

Table Q4(c): Cost breakdown for a transceiver

Cost Category	Cost per Transceiver (RMB)
Direct Materials	22.50
Direct Labour	12.50
Variable Overhead	2.50

Given this new information, prepare a new cost table showing the annual impact of manufacturing the transceivers within the production facilities available in the company. Should your company continue purchasing the transceivers from the external supplier or start manufacturing their own transceivers? [5]

(d) Using the information provided in **Table Q4(c)**, if your company can sell the optical transceivers for 78.125 RMB each, draw a table showing the production costs and sales revenue of manufacturing and selling transceiver quantities from 3,500 → 17,500 transceivers in 3,500 transceiver increments. [5]

(e) If the factory is running at full capacity of 17,500 transceivers per year with an assumed price of 78.125 RMB per transceiver, what is the profit made by the company? What is the safety margin for this scenario? [3]

(f) Draw a Breakeven graph showing Total Cost (**TC**), Variable cost (**VC**), Fixed Cost (**FC**), and Revenue (**REV**) against production quantity (**Q**). Calculate the Breakeven Point and indicate this on the graph. [5]

SAMPLE SET 4

Section A: Multiple Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options to each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) Which item below is correct regarding Design for Manufacturing (DFM): [2]

F. As a general rule, DFM can't be started at the initial stages of product design before fabrication.

G. It is usually accepted (demonstrated) that 70 to 80% of production costs are determined by design decisions.

H. DFM is a design method to reduce the overall cost of production by increasing the complexity of manufacturing operations.

I. The DFM may impose higher manufacturing costs, but instead, it can benefit from higher customer satisfaction because of improved product quality.

J. While manufacturers were facing strong competition from other manufacturers (rivals), at the time, they were under pressure to increase production efficiency while increasing costs of production.

(1.2) Which statement accurately describes a cause-effect diagram? [2]

F. A cause-effect diagram is primarily used to illustrate the effects of a specific problem without identifying its root causes.

G. Cause-effect diagrams, also known as fishbone or Ishikawa diagrams, are visual tools used to focus on the sole cause of a given problem.

H. A cause-effect diagram called the fishbone or Ishikawa diagram is designed to indicate the relationships between various effects easily with no need for visual tools.

I. A cause-effect diagram is a visual tool used to logically organize possible causes for a specific problem or effect by graphically displaying them in increasing detail, suggesting causal relationships among theories.

J. Cause-effect diagrams called fishbone or Ishikawa diagrams are visual tools exclusively used to identify effects, without the need for organizing possible causes or establishing causal relationships.

(1.3) Which of the following statements best describes the three pillars of sustainability? [2]

K. Sustainability involves meeting the current generation's needs without consideration for future generations, emphasizing economic growth, environmental care, and social well-being.

L. Sustainability entails achieving economic growth, environmental care, and social well-being simultaneously, fulfilling the needs of current generations without compromising the needs of future generations.

M. Sustainability primarily focuses on economic growth, with minimal consideration for environmental and social aspects, to fulfill the needs of the current generation.

N. Sustainability prioritizes environmental care over economic growth and social well-being, neglecting the importance of balancing all three pillars.

O. Sustainability emphasizes social well-being as the sole pillar, with economic growth and environmental care being secondary considerations in fulfilling the needs of current and future generations.

(1.4) Which of the following is not a project? [2]

K. Programming regular software updates for an electronic control system in a manufacturing plant.

L. Developing and implementing a software algorithm for real-time data processing in an electronic control system

M. Developing a prototype for a next-generation smartphone.

N. Implementing a recycling program in an office building.

O. Designing and building a printed circuit board (PCB) for a new electronic device.

(1.5) Which of the following is not true about the initial phase of a project? [2]

F. The cost associated at the beginning of the project is highest.

G. Stakeholders have the maximum influence during this phase.

H. The highest uncertainty is at this stage of the project.

I. The initial scope is defined, and initial financial resources are committed.

J. All the above statements are correct.

(1.6) Which of the following statements about the WBS is most correct in the project management context? [2]

A. WBS stands for Work Breakdown Statement.

B. The WBS includes 100 percent of the work of the project.

C. Any work not included in the WBS is out of scope.

D. WBS can decompose through project work packages and cannot by phases of a project.

E. B & C

(1.7) Consider the structure of a start-up company run by an entrepreneur. Find the correct pair of true statements. [2]

5 Maximum standardisation and formalisation
6 Few layers: Limited middle-line managerial levels
7 Decentralised and indirect supervision
8 Wide span of control (around the entrepreneur)

A – 1 and 2

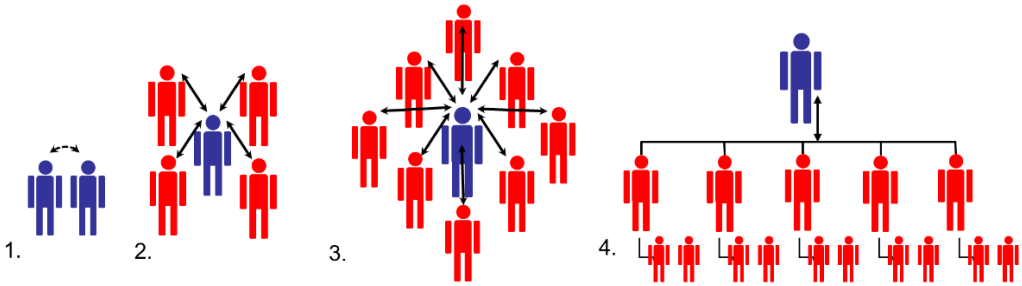
B – 3 and 4

C – 2 and 3

D – 2 and 4

E – 1 and 3

(1.8) The following organisational structures should be labelled as: [2]



A – 1. Start-up; 2. Expansion; 3. Growth; 4. Formal Organisation

B – 1. Expansion; 2. Start-up; 3. Formal Organisation; 4. Growth

C – 1. Start-up; 2. Growth; 3. Expansion; 4. Formal Organisation

D – 1. Expansion; 2. Start-up; 3. Growth; 4. Formal Organisation

E – 1. Growth; 2. Start-up; 3. Formal Organisation; 4. Expansion

(1.9) It costs 1000 RMB for hand tools and 1.50 RMB labour per unit to manufacture a product. An alternative is to manufacture the product with automated machinery costing 15,000 RMB, with a 0.50 RMB per-unit cost. With an annual production rate of 5000 units, how long will it take to reach the production cost cross-over point? [2]

A – 2.0 yr

B – Never

C – 3.6 yr

D – 2.8 yr

E – 0.9 yr

(1.10) A company has been in business for approximately 12 months and has prepared financial projections to present to an investor. Over the last 12 months the company has sold **300,000 CNY** of their existing products which cost them **150,000 CNY** to manufacture. In addition, the company has been working on a new product design which cost **100,000 CNY** to develop and has been valued at **200,000 CNY**. The new product will replace the existing product so all existing product designs are valued at **0 CNY**. What is the present valuation of the company? [2]

A – 250,000 CNY

B – 150,000 CNY

C – 500,000 CNY

D – 50,000 CNY

E – 350,000 CNY

(1.11) Project risk management is best described as: [2]

A – Managing responses to threats

B – Identifying and acknowledging threats and opportunities

C – Planning responses to threats

D – Minimising threats and maximising opportunities

E – Implementing responses to threats

(1.12) Risk response planning consists of four main strategies. Which of the following does not fall under the main strategies of risk response planning? [2]

A – Avoidance

B – Acceptance

C – Transference

D – Mitigation

E – Identification

Section B: Long Questions

Q2 As a young entrepreneur you would like to guarantee a healthy total cost of ownership via evaluating net present value of your young company while utilising discounted cash flow analysis.

(a) Consider your company is designing a new product that will take your company three years to design, fabricate, and build. For this product, your company's expenses are 1,000,000 RMB per year, which will increase with inflation rate of 5% annually. How much must your company quote your customer (for delivery in three years time) if your company would like to cover their expenses and earn a net profit (in today's equivalent currency) of 100,000 RMB? [3]

(b) Consider your company would like to purchase a new machine that has a purchase price of 10 Million (M) RMB with a lifetime of five years. The machine will require annual maintenance costs of 1 M RMB associated with annual power costs of 0.5 M RMB. All costs are fixed with inflation of 10% applied to power only after year 1.

(i) Demonstrate the total projected costs over the lifetime of this new machine. [3]

(ii) Analyse the total cost of ownership over the lifetime of this new machine. [4]

(c) Consider you have decided that your company will make a significant investment into expanding its presence in China by setting up a large assembly facility in Chengdu. Your company has estimated its initial set up costs to be in the region of 6,398 M RMB. Net income from this new assembly facility is forecasted as follows:

Year (Y) 1: 1,400 M RMB
Y2: 1,450 M RMB
Y3: 1,550 M RMB
Y4: 1,625 M RMB
Y5: 1,480 M RMB

(i) Calculate the projected payback time for this new assembly facility to the nearest month. [5]

(ii) Calculate the net present value of this new assembly facility based on a discounting factor of 5%. [5]

(iii) Comment on the attractiveness of this new assembly facility. [3]

(d) Consider a part of cash-flow with 30 days net payment terms for cash-in and cash-out given in Table Q2 below. Determine the cumulative cash towards the

end of each month (M). The currency utilised for transactions in this cash-flow is RMB. [3]

Table Q2: Cash-Flow Table

	M0	**M1**	**M2**	**M3**	**M4**	**M5**
Sales Bookings	10	12	8	20	100	0
Shipments	0	10	12	8	20	100
Components Order	5	6	4	10	50	0

Q.3

(l) DFM is the process of designing components for ease of manufacturing high-quality products at a lower cost.

(vi) The product development cycle in general consists of the initial design, final design, fabrication, production, and product launch. Draw a plot while showing the effectiveness of the DFM application versus the product development cycles (aforementioned) based on both the impact and cost of changes. [4]

(vii) Sustainability can make a balance between three main pillars. List any two of these pillars. [2]

(m) The goal of Quality control (QC) is to identify any defects after a product is developed, but before it's released to production. By using seven basic tools of quality, we can manage the quality of our product or process effectively, known around the world as the seven quality control (7-QC) tools:

(iv) Which 7-QC tool looks at a bar chart that is sorted in descending order from the highest frequency to the lowest frequency from left to right, while the height of the bars reflects the frequency or the impact of the problems? [1]

(v) Which of the 7-QC tools is a graph used to study how a process changes over time? [1]

(vi) Assume fifteen sample points are collected for yield rate over time in a specific process, see Table Q3(b). Utilizing the tool mentioned in (ii), create a plot demonstrating an uncontrolled variation, indicative of an unstable process. Provide a visual representation by including a central line, upper and lower

control limits, and notes highlighting the instability of the process. [6]

Sample Number	Yield Rate (%)
1	98.50
2	97.80
3	98.30
4	99.10
5	99.30
6	94.32
7	98.40
8	98.60
9	98.80
10	99.00
11	99.10
12	98.80
13	98.90
14	99.10
15	98.20

Table Q3(b)

(n) Robust Design is an engineering methodology for improving productivity during research and development so that high-quality products can be produced quickly and at a low cost.

(iii) Consider a power supply circuit used in a stereo system for which the target output voltage is 110 volts. If the output voltage falls outside 110±20 volts, then the stereo fails in half the situations and must be repaired. Suppose it costs $100 to repair the stereo. Using the Taguchi quadratic loss function, compute the average loss associated with a particular output voltage of 100 volts. [3]

(iv) How can the concept of robust design increase the quality of a process/product (hint: the main idea behind the robust design)? [2]

(o) In an electronic manufacturing process, a new component is designed with a parameter range of 20±1 mm. The process, characterized by a mean value of 19.8 mm and a standard deviation of 0.2 mm, is assessed for its capability to meet the specifications. Answer the following questions:

(i) Calculate the process capability index Cpk for the given parameter range. [4]

(ii) Discuss your assessment over the obtained Cpk value in term of the risk associated with this manufacturing process. [2]

Q4

(c) Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements. Answer the following questions.

(v) Project A has a BCR of 1.4. What can we infer from this? [2]

(vi) During one of the standup meetings, a team member asked you the difference between standard and regulation. How would you respond? [2]

(vii) Compare the main elements involved in "Sample Basic Sender / Receiver" with Interactive" communication models in a project. [4]

(d) Project schedule management includes the processes required to manage the timely completion of the project. One of the common techniques used is CPM (critical path method). The critical path is the sequence of activities that represents the longest path through a project, which determines the minimum possible project duration. Consider a project (assume date 0 for project starting) with the details shown in Table Q4(b) and answer the following questions.

Activity	Duration	Predecessor(s)
A	2	-
B	1	-
C	4	-
D	1	A
E	2	B,C,D
F	7	C

Table Q4(b)

(v) Plot a CPM network diagram for the above project (hint: show activity on nodes and concerning their relevant dependencies). [4]

(vi) Perform the forward pass analysis through the schedule network and compute Early Start and Early Finish dates for all activities. [4]

(vii) Perform the backward pass analysis through the schedule network and compute Late Start and Late Finish dates for all activities. [4]

(viii) Calculate the total float (slack) for all activities. [2]

(ix) Define the Critical Path (sequence of activities that are located on the critical path). [1]

Assume the activity "C" is completed with 1 day delay. Discuss how the consequence of this delay can affect the project. [2]

SAMPLE SET 5

Section A: Multiple-Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options for each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) What is a critical path? [2]

A – The longest path through the project network with the least amount of slack

B – The shortest path through the project network with the maximum amount of slack

C – The shortest path through the project network with the least amount of slack

D – The shortest chain of tasks that considers both task dependencies and resource dependencies

E – The longest chain of tasks that considers both task dependencies and resource dependencies

(1.2) Consider the structure of a start-up company run by an entrepreneur. Find the correct pair of true statements. [2]

9 Maximum standardisation and formalisation
10 Few layers: Limited middle-line managerial levels
11 Decentralised and indirect supervision
12 A wide span of control (around the entrepreneur)

A – 1 and 2

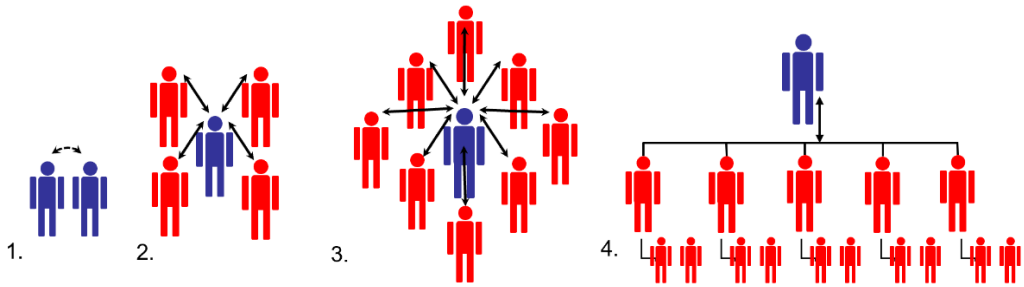
B – 3 and 4

C – 2 and 3

D – 2 and 4

E – 1 and 3

(1.3) The following organisational structures should be labelled as: [2]



A – 1. Start-up; 2. Expansion; 3. Growth; 4. Formal Organisation

B – 1. Expansion; 2. Start-up; 3. Formal Organisation; 4. Growth

C – 1. Start-up; 2. Growth; 3. Expansion; 4. Formal Organisation

D – 1. Expansion; 2. Start-up; 3. Growth; 4. Formal Organisation

E – 1. Growth; 2. Start-up; 3. Formal Organisation; 4. Expansion

(1.4) A project has more than one critical path. What does this mean? [2]

A – Risk increases

B – Risk decreases

C – The duration of the project reduces

D – The project may cost less

E – Nothing happens

(1.5) Which document is created by breaking a project scope into smaller and more manageable elements? [2]

A – Scope baseline

B – Organisational breakdown structure (OBS)

C – Work breakdown structure (WBS)

D – Statement of work (SOW)

E – Budget report

(1.6) In the execution phase of a project, key resources are lost, and new staff joins. The project manager sets up a meeting with the new team. What should be the most important agenda item for this meeting? [2]

A – Discussing team building activities

B – Reviewing WBS and responsibilities of all team members

C – Inviting the team members to share their past project experiences

D – Planning risk responses for existing risks

E – All of the above

(1.7) It costs 1000 RMB for hand tools and 1.50 RMB labour per unit to manufacture a product. An alternative is to manufacture the product with automated machinery costing 15,000 RMB, with a 0.50 RMB per-unit cost. With an annual production rate of 5000 units, how long will it take to reach the production cost cross-over point? [2]

A – 2.0 yr

B – Never

C – 3.6 yr

D – 2.8 yr

E – 0.9 yr

(1.8) Design for Manufacturing (DFM) is an important part of any product development cycle. Which item below is not correct regarding DFM: [2]

A – The time required for design for manufacturability is determined by the product's complexity, the organization's size, and the resources available.

B – As a general rule, DFM is a continuous process that begins at the initial stages of product design and continues through the entire product lifecycle.

C – The DFM is more effective to be applied to the initial design because of the highest impact and lowest cost of changes.

D – The DFM may impose higher manufacturing costs, instead can benefit from higher customer satisfaction because of improved product quality.

E – The DFM consists of some main principles such as Simplicity, Standardization, Tolerance, Material Selection, Automation, and Process Integration.

(1.9) In the development of a new product, the impact on Price, Quality, and Manufacturing time is: [2]

A – Product Development: 20-30%, Manufacturing: 70-80%

B – Product Development: 70-80%, Manufacturing: 20-30%

C – Product Development: 50%, Manufacturing: 50%

D – Sales & Marketing: 33.3%, Product Development: 33.3%, Manufacturing: 33.3%

E – Sales & Marketing: 50%, Product Development: 50%

(1.10) Which of the following statements best fits with the definition of the control charts? [2]

A. Control charts are used to determine whether a process is stable and predictable over time.

B. Control charts are used to determine whether a process is profitable or not.

C. Control charts are used to determine whether a process is meeting customer demand or not.

D. Control charts are used to determine whether a process is efficient or not.

E. None of the above statements is correct.

(1.11) Which of the following statements is true regarding process capability? [2]

A. Process capability is a measure of how efficient a process is in terms of time and resources.

B. Process capability refers to the degree to which a process can consistently produce output that meets process specifications.

C. Process capability is determined solely by the skill level of the workers involved in the process.

D. Process capability is not relevant for service industries.

E. None of the above statements is true regarding process capability.

(1.12) The three variables of the project planning process: [2]

A – Cost, Time, Performance

B – Price, Delivery, Quality

C – Power, Speed, Cost

D – Slack Time, Tasks, Resources

E – Manpower, Critical Path, Temperature

Section B: Long Questions

Q2 As a young entrepreneur you would like to guarantee a healthy total cost of ownership via evaluating the net present value of your young company while utilising discounted cash flow analysis.

(a) Consider your company is designing a new product that will take your company three years to design, fabricate, and build. For this product, your company's expenses are 1,000,000 RMB per year, which will increase with an inflation rate of 5% annually. How much must your company quote your customer (for delivery in three years' time) if your company would like to cover their expenses and earn a net profit (in today's equivalent currency) of 100,000 RMB? [3]

(b) Consider your company would like to purchase a new machine that has a purchase price of 10 Million (M) RMB with a lifetime of five years. The machine will require annual maintenance costs of 1 M RMB associated with annual power costs of 0.5 M RMB. All costs are fixed with inflation of 10% applied to power only after year 1.

(i) Demonstrate the total projected costs over the lifetime of this new machine. [3]

(ii) Analyse the total cost of ownership over the lifetime of this new machine. [4]

(c) Consider you have decided that your company will make a significant investment into expanding its presence in China by setting up a large assembly facility in Chengdu. Your company has estimated its initial setup costs to be in the region of 6,398 M RMB. Net income from this new assembly facility is forecasted as follows:

Year (Y) 1: 1,400 M RMB
Y2: 1,450 M RMB
Y3: 1,550 M RMB
Y4: 1,625 M RMB
Y5: 1,480 M RMB

(i) Calculate the projected payback time for this new assembly facility to the nearest month. [5]

(ii) Calculate the net present value of this new assembly facility based on a discounting factor of 5%. [7]

(iii) Comment on the attractiveness of this new assembly facility. [3]

Q.3

(p) DFM is the process of designing components for ease of manufacturing high-quality products at a lower cost.

(viii) The product development cycle in general consists of the initial design, final design, fabrication, production, and product launch. Which stage is more effective in DFM regarding the impact and cost of changes? [3]

(ix) Sustainability can make a balance between three main pillars. List any two of these pillars. [2]

(q) The goal of Quality control (QC) is to identify any defects after a product is developed, but before it's released to production. By using seven basic tools of quality, we can manage the quality of our product or process effectively. Known around the world as the seven-quality control (7-QC) tools.

(vii) List any two of these tools. [2]

(viii) A company that produces electronic devices is experiencing a high rate of product returns due to defects. How might a cause-effect diagram be used to identify the root causes of these defects? [4]

(ix) The cost of quality can be categorized into four categories including Prevention Cost, Appraisal Cost, Internal Failure Cost, and External Failure Cost. Classified two types of costs accordingly, first the cost of quality-related training, and second product recall and warranty repairs. [2]

(r) Robust Design is an engineering methodology for improving productivity during research and development so that high-quality products can be produced quickly and at a low cost.

(v) From the viewpoint of robust design, an engineering design problem can be optimized through three main steps. List any two of these steps. [2]

(vi) The response of the product is denoted by y. The response can be the output of the product or some other suitable quality characteristic. Three classes of parameters (as shown in Figure Q3(c) by Z, M, and X) can influence the response or quality characteristic of the product. List any two of them. [2]

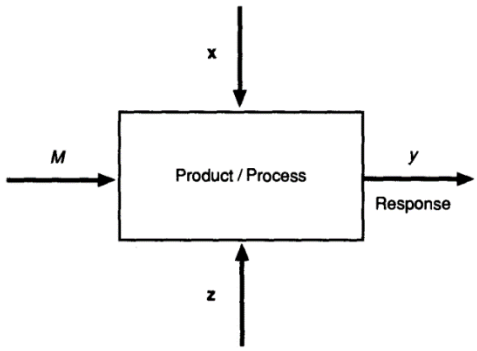


Figure Q3(c)

(vii) Assume the product engineer collect 10 observation (test results) from one product's quality characteristic (nominal the best). If the target point for this quality characteristic is 8.5, compute the Taguchi loss function for this set of data (assume constant $k = 1$ is the associated loss function formulation). [3]

Observation No. 1 = 8.100

Observation No. 2 = 8.900

Observation No. 3 = 8.450

Observation No. 4 = 9.250

Observation No. 5 = 8.860

Observation No. 6 = 8.350

Observation No. 7 = 8.250

Observation No. 8 = 8.680

Observation No. 9 = 8.900

Observation No. 10 = 9.050

(s) Six-sigma is a statistical concept that measures a process in terms of defects – at the six sigma level, there are 3.4 defects per million opportunities:

(iii) What is Critical-to-Quality (CTQ)? [2]

(iv) A manufacturing process produces a new product with an upper and lower specified length of 20 ± 1 mm. The process has a mean diameter of 19.8 mm and a standard deviation of 0.2 mm. Calculate the process capability index Cpk. [3]

Q4

(a) Derive the theoretical break-even point relationship in terms of fixed costs, variable costs, and selling price. [3]

You work for ABC manufacturing company that supplies the industry with Wi-Fi routers. The company has an annual production capacity of 45,000 routers. Your production cost breakdown for 45,000 routers is shown in Table Q4 below.

Table Q4: Cost breakdown for 45,000 routers

Cost category	Annual Cost (RMB)
Direct Materials	405,000
Direct Labour	225,000
Variable Overhead	45,000
Fixed Overhead Cost	562,500
Total	**1,237,500**

An outside supplier has approached your company and offered to supply the Wi-Fi routers to your company for 27.5 RMB each. Your manager has asked you to carry out an analysis of this offer. He advises you that the Fixed Overhead Costs for the production facilities will NOT be avoided if you purchase the routers from this outside supplier as your company has no other use for the vacant production facilities if you purchase from the outside supplier.

(b) Draw a table comparing the cost of making the routers internally on a per router basis with the costs of purchasing the routers from the outside supplier. Do you recommend that your company purchases from an outside supplier? [5]

(c) Your manager now tells you he can rent out the production facilities for 570,000 RMB per year. Given this new information, prepare a new cost table showing the impact of the facilities rental. Does this information change your recommendation on whether to purchase from an outside supplier? [5]

(d) Utilising the information provided in **Table Q4**, if your company can sell the Wi-Fi routers for 40.00 RMB each, draw a table showing the production costs and sales revenue of manufacturing and selling router quantities from 10,000 → 30,000 routers in 5000 router increments. [5]

(e) If the factory is running at full capacity of 45,000 routers per year with an assumed price of 40.00 RMB per router, what is the profit made by the company? Also, calculate the safety margin for this scenario. [3]

(f) Draw a Breakeven graph showing Total Cost (**TC**), Variable cost (**VC**), Fixed Cost (**FC**), and Revenue (**REV**) against production quantity (**Q**). Calculate the Breakeven Point and indicate this on the graph. [5]

SAMPLE SET 6

Section A: Multiple Choice Questions

Q1 Multiple Choice Questions (MCQs): There are 12 MCQs. You are provided with five options to each question (A, B, C, D, or E). ONLY one of the five is the correct answer for each question.

(1.1) Design for Manufacturing (DFM) is an important part of any product development cycle (e.g., initial design, final design, fabrication, production, and product launch). Which item below is correct about when to apply DFM in the initial design or production stage? [2]

F. Applying the DFM in an initial design stage provides a higher impact with lower cost compared with the production stage.

G. Applying the DFM in an initial design stage provides a higher impact but higher cost compared with the production stage.

H. Applying the DFM in a production stage provides a higher impact with lower cost compared with the initial design stage.

I. Applying the DFM in a production stage provides a higher impact but higher cost compared with the initial design stage.

J. Applying the DFM in both stages can provide the same impact with the same amount of required cost.

(1.2) There is a variety of actionable sustainable design principles that can be implemented to create better designs. Some of them are below except: [2]

F. Dematerialization.

G. Migration to product-service systems.

H. Limit or eliminate long-distance outsourcing.

I. Design for a shorter period of usage.

J. Invest in simulation.

(1.3) The cost of quality can be categorized into four categories: Prevention Cost, Appraisal Cost, Internal Failure Cost, and External Failure Cost. Which item below is not classified as an appraisal cost? [2]

A. Test and inspection (receiving, in-process, and final)

B. Supplier acceptance sampling

C. Product Audits

D. Equipment upgrades

E. Calibration

(1.4) Project A has an NPV of $1.2 million; Project B has a BCR of 1.035; Project C has an IRR of -0.03 percent. Which of these projects appears to be most attractive? [2]

F. Project A is the most attractive.

G. Project B is the most attractive.

H. Project C is the most attractive.

I. All three projects A, B, and C are most attractive equally to execute.

J. It is uncertain which project is the most attractive to execute based on the provided information.

(1.5) SWOT is a technique used in project management, mainly for which item below? [2]

F. Silver Whisker Optimization Technique.

G. A tool or technique used in identifying risks.

H. A tool or technique for structured ranking of cusses/effects.

I. A graphic view of relative probabilities.

J. None of the above.

(1.6) Silver Springs Financial Services is migrating servers as part of a data center exit project, triggered by a need to slash costs - the fallout of a recent rogue trading fiasco. The project manager's EVM report today shows a CPI of 1.1 and an SPI of 0.9. What should we understand from this? [2]

F. Trading controls need to be tightened.

G. The project is under budget and behind schedule.

H. The project is over budget.

I. The project is ahead of schedule.

J. Choices of C&D

(1.7) In a Profit and Loss (Income) statement, Earnings before Interest, Tax, Depreciation, and Amortisation (EBITA) is equal to: [2]

A – Total Sales minus Cost of Sales

B – Total Expenses minus Net Profit

C – Total Sales minus (Salaries + Rent)

D – Gross Profit minus all company operating expenses (salaries and rent etc.)

E – Full production costs minus Net Profit

(1.8) If a company has an organisational structure as shown below in Figure Q1(1.3), it is called: [2]

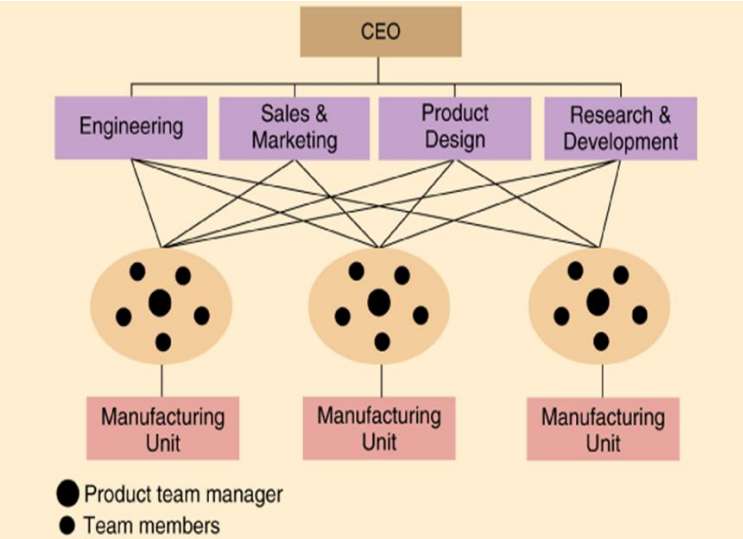


Figure Q1(1.3)

A – A Team-Based Structure

B – A Matrix Structure

C – A Divisional Structure

D – A Simple Startup Structure

E – A Military Command Structure

(1.9) A Sole Trader Company is: [2]

A – A business you run as an individual. You can keep all your business' profits after you have paid tax on them. You are personally responsible for any losses your business makes.

B – A business in which you and your business partner (or partners) personally share responsibility for your business. You can share all your business's profits between the partners. Each partner pays tax on their share of the profits.

C – A company whose shares can be freely traded on the stock market.

D – An organisation you can set up to avoid paying tax or having any responsibility as a director.

E – An organisation you can set up to run your business- it is responsible for everything it does, and its finances are separate from your personal finances. Any profit it makes is owned by the company after it pays Corporation Tax. The company can then share its profits amongst its shareholders.

(1.10) In a simple Lease–or-Buy calculation, you require a radar equipment for a project. The cost to buy the equipment is 30,000 RMB plus an additional 500 RMB per day. You are able to lease the equipment from a local company at a cost of 2000 RMB per day. What is the maximum length of time you can lease the equipment before it is cheaper to buy the equipment? [2]

A – 3 days

B – 10 days

C – 20 days

D – 30 days

E – 16.333 days

(1.11) What is a Profit & Loss (Income) statement? [2]

A – It only summarises all the sales revenues for the financial year

B – It only summarises all the payments or expenses for the financial year

C – It is only used when the company makes a loss

D – It summarises the sales revenues, payments, and expenses for the financial year

E – It is only used when the company makes a profit

(1.12) If a company is described as having a 'Defender (i.e. market leader or stable market)' Strategic Posture, it displays the following characteristics: [2]

	Emphasis	**Market Focus**	**Structure and Process**	**Technology**
A -	Pioneering changes and innovation	Aggressively explore and exploit new market opportunities	Organic: informal, decentralised, and flexible	Contingent, multiple, prototypical technologies; low degree of routinisation and mechanisation
B -	Stability and efficiency	Maintain a small niche	Mechanistic: hierarchical, centralised, and control	One core highly cost-efficient technology
C -	Balanced: minimised risks and maximise profits	Defend core business while cautiously exploit proven new markets by imitating Prospectors	Matrix organisation – coalition between core functions	Dual core with emphasis on applied research
D -	Reluctant to act due to instability and uncertainty	Unsure	Maintain whatever structure is in place; often a weak link with strategy, which is unclear	Unsure
E -	Stability and efficiency	Aggressively explore and exploit new market opportunities	Organic: informal, decentralised, and flexible	Dual core with emphasis on applied research

Section B: Long Questions

Q2

(a) You are a project manager working on China's high-speed railway. You need to install a major new piece of electrical equipment into a new station that is being built at Chengdu's new airport. The only way to install the new equipment is to hire a tower crane to lift it into position. As the project manager, you have collected all the information and tasks required to build a tower crane on the station site and recorded the tasks in Table Q2(a).

Table Q2(a): Tower Crane tasks and durations.

Task Number	Task Description	Duration (days)	Predecessors
0	Project Start (Milestone)	0	
1	Specify Crane	3	0
2	Survey Site	4	1
3	Survey Power Supply	3	1
4	Select Site	1	2
5	Order Tower Crane	2	3,4
6	Arrange Transport	4	5
7	Order Mobile Erector Crane	3	5
8	Install Concrete Plinth	5	5
9	Confirm Site Access	1	6,7,8
10	Get Mobile Erector Crane on Site	4	9
11	Crane Parts on Site	5	9
12	Install Crane Base	2	10,11
13	Install Tower & Crane Head	1	12
14	Assemble Jib	2	10,11
15	Install Jib	1	13,14
16	Install counterweight	1	15
17	Apply Crane Power	1	16
18	Test Crane	1	17
19	Project end	0	18

Show the above information on a Network diagram showing each task duration and the interdependencies of each task. [8]

(b) Using the obtained Network diagram:

(ii) List the critical path tasks in the order they occur. [3]

(ii) Calculate the project's overall duration (show your working). [2]

(iii) If Task 8 (Install Concrete Plinth) cannot be started until the 'Mobile Erector Crane' has been ordered (Task 7) has been completed. Calculate the new project duration and describe how this change impacts the critical path. [2]

(c) Figure Q2(c) shows five process groups (developed by PMI) that lead to a systematic approach to project management including initiating processes, planning processes, executing processes, controlling processes, and closing processes. Assume you are a member of a student team developing a mobile study app, answer the following questions accordingly.

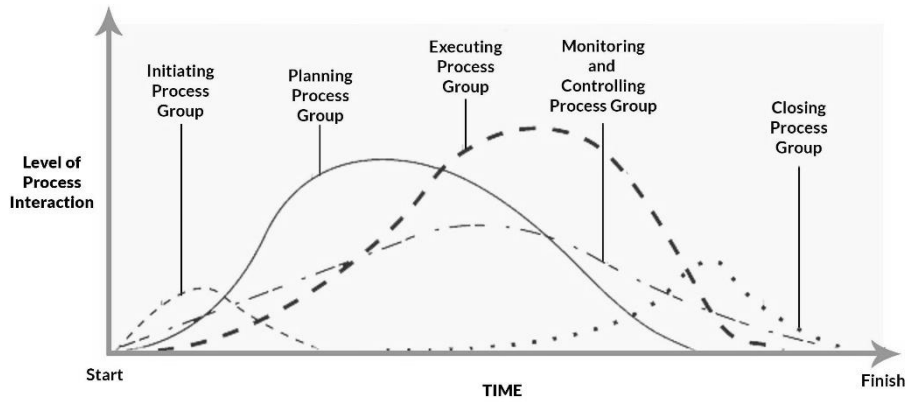


Figure Q2(c)

(i) At this time, the team is managing changes and recommending actions as needed, overseeing project activities against the plan, and ensuring only approved modifications are implemented. In which process group is the team currently operating? [1]

(ii) At this time, the team outlines strategies and tactics for project completion, identifies potential risks, and documents the execution processes for the project. In which process group is the team currently operating? [1]

(iii) At this time, the team clarifies the project's purpose and scope to the stakeholders. In which process group is the team currently operating? [1]

(iv) At this time, the team is ensuring all project tasks are completed across different stages, obtaining approval from stakeholders to formally conclude the project phase, and updating project documents as necessary for future reference. In which process group is the team currently operating? [1]

(d) Leadership and Management are two concepts in project management that can differ in several important aspects. Identify and compare three main differences between these two concepts. [3]

(e) In a project, the Planned Value (PV) is $10,000, the Earned Value (EV) is $8,500, and the Actual Cost (AC) is $8,000.

(i) Calculate the Schedule Performance Index (SPI) for this project. [1]

(ii) Calculate the Cost Performance Index (CPI) for this project. [1]

(iii) Based on the obtained SPI and CPI, what do you infer regarding the performance of this project so far? [1]

Q3

(t) Design For Manufacturing (DFM) consists of some main principles such as Simplicity, Standardization, Tolerance, Material Selection, Automation, and Process Integration.

(i) In the context of Six Sigma methodologies, explain how Poka-yoke can contribute to process improvement. Provide a specific example of a Poka-yoke mechanism that could be implemented in a laboratory setting involving the use of LEDs. [4]

(ii) A manufacturer is producing a specific type of resistor with a target resistance of 12 ohms and a tolerance of ±2 ohms. To assess the manufacturing process's capability to meet specifications, the manufacturer measures the resistance of a sample of resistors and computes the Cp and Cpk indices. Given that the sample of resistors has an average resistance of 12.2 ohms and a standard deviation of 0.5 ohms, evaluate both process capability indicators for this manufacturing operation. [4]

(iii) The 4Rs are highly effective tools in establishing a circular economy for sustainable products. Comprising four principles applicable to almost every individual, these principles have an exponential impact. Name any two of these principles. [2]

(u) Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.

(iv) Project managers should have three key skill sets (Talent Triangle). Select and explain briefly any two of these skills. [4]

(v) What is the term used to describe the process of developing a detailed description of the project and product, often represented as a hierarchical decomposition of the project scope? [1]

(vi) Sarah is overseeing the development of a new electronic device prototype. After collaborating with the team, they arrive at a most likely estimate of 12

weeks for the total project duration. In the best-case scenario, they anticipate a potential reduction of three weeks due to efficient collaboration with a specialized component supplier. However, there's a risk of a delay in obtaining critical components that could extend the project pessimistically by four weeks. Using three-point estimating, calculate the expected duration based on Triangular Distribution and Beta Distribution separately, and determine the standard deviation of these estimates. [4]

(vii) You are managing a team of 6 members and once the project enters the next phase, three more members are added and two of the current team members leave. How many new communication channels need to be added due to these changes? [3]

(viii) In conflict management, understanding various techniques is crucial, e.g., Withdraw/Avoid, Smooth/Accommodate, Compromise/Reconcile, Force/Competing, and Collaborate/Problem Solve. Among the aforementioned techniques identify which method raises a win-win situation where both sides benefit, which results in a lose-lose scenario for both parties, and which method leads to a situation where neither side gains nor loses? [3]

Q4

(a) Derive the theoretical break-even revenue relationship in terms of fixed costs, variable costs, and selling price. [3]

You work for XYZ, a company that requires 17,500 optical transceivers annually to manufacture its optical wireless communication (OWC) systems. These optical transceivers are supplied by an external supplier to your company for 57 RMB each. Moreover, your company has some vacant production facilities that adds to the total annual costs with Fixed Overhead Cost of 568,750 RMB. Currently, the vacant production facilities are rented out for 225,000 RMB per year but your manager has informed you that the rental agreement is nearing an end soon.

(b) Draw a table comparing the equivalent cost of purchasing a transceiver from the external supplier while the vacant production facilities are rented out with the scenario when the rental agreement has expired. [5]

(c) Your manager now tells you he wants to utilise the vacant production facilities and manufacture the transceivers within the company. In addition to the Fixed Overhead Cost, the production cost breakdown for a transceiver is shown in Table Q4(c) below.

Table Q4(c): Cost breakdown for a transceiver

Cost Category	Cost per Transceiver (RMB)
Direct Materials	22.50
Direct Labour	12.50
Variable Overhead	2.50

Given this new information, prepare a new cost table showing the annual impact of manufacturing the transceivers within the production facilities available in the company. Should your company continue purchasing the transceivers from the external supplier or start manufacturing their own transceivers? [5]

(d) Using the information provided in **Table Q4(c)**, if your company can sell the optical transceivers for 78.125 RMB each, draw a table showing the production costs and sales revenue of manufacturing and selling transceiver quantities from 3,500 → 17,500 transceivers in 3,500 transceiver increments. [5]

(e) If the factory is running at full capacity of 17,500 transceivers per year with an assumed price of 78.125 RMB per transceiver, what is the profit made by the company? What is the safety margin for this scenario? [3]

(f) Draw a Breakeven graph showing Total Cost (**TC**), Variable cost (**VC**), Fixed Cost (**FC**), and Revenue (**REV**) against production quantity (**Q**). Calculate the Breakeven Point and indicate this on the graph. [5]

Sample Exam Questions

Engineering Project Management & Finance

红字答案解析 Annotated Solutions

说明: 前 54 页保留原题 PDF；后续部分按 Sample Set
和题号给出中文解析，专有名词保留英文标注。

Sample Set 1

Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	DFM 正确描述	D。DFM (Design for Manufacturing) 目标是降低制造复杂度和总生产成本。
1.2	Sustainable design principles except	D。Design for a shorter period of usage 与可持续设计相反。
1.3	7-QC 组织原因关系	A。Cause-Effect Diagram (Fishbone/Ishikawa) 用来组织可能原因。
1.4	Six Sigma roadmap	A。DMAIC = Define, Measure, Analyze, Improve, Control。
1.5	NPV/BCR/IRR 项目吸引力	E。NPV、BCR、IRR 口径不同，不能仅凭这些异质指标直接比较三项目。
1.6	成员缺技能但自认专家	D。Unconsciously incompetent: 不知道自己不知道。
1.7	EBITDA	D。EBITDA = Gross Profit - operating expenses。
1.8	电池健康风险、不确定证据	B。采用 precautionary approach，增加防护和泄漏监测。
1.9	组织结构图	B。图中同时有 division 与 function 双线结构，是 Matrix Structure。
1.10	无议程会议混乱	B。Meeting agenda 是恢复会议秩序的首要工具。
1.11	30 天账期现金流	A。累计现金为 M0=0, M1=-5, M2=-1, M3=7, M4=5, M5=-25 RMB。
1.12	现金流性质	C。销售增长导致现金短缺，属于 over trading。

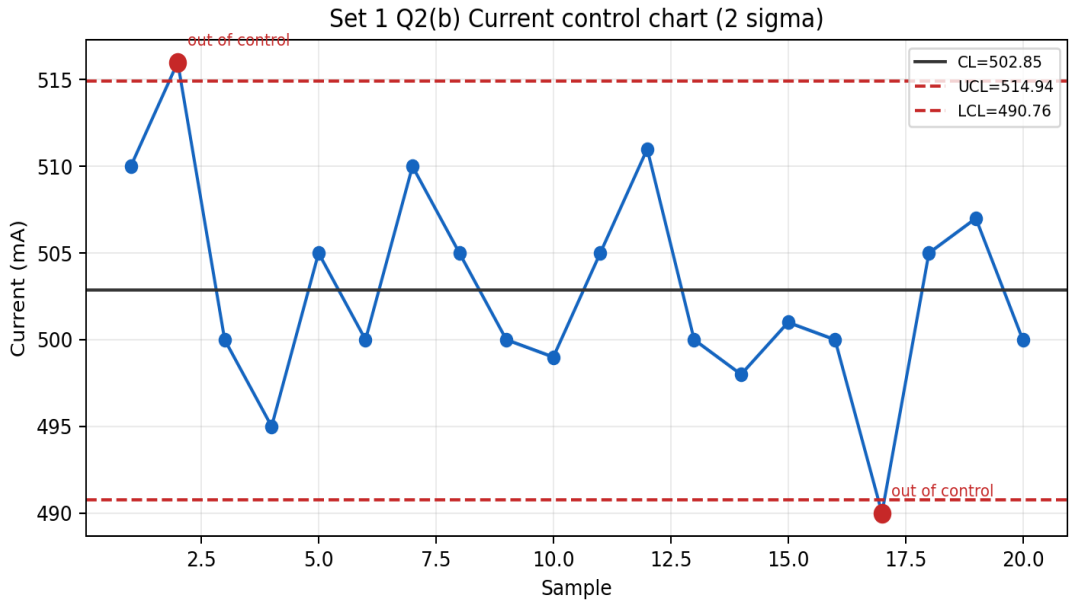
Q2 DFM, Sustainability, Control Chart, Process Capability

原题 Q2(a): Design A 为 5 个独立零件，Design B 为单一 molded component。

答案解析: Design A 的 DFM 优点是模块化、可使用标准件、可维修和可复用；缺点是装配步骤多、人工和夹具成本高、库存和公差链更复杂。Design B 的优点是零件数少、装配时间短、尺寸小、潜在可靠性更高；缺点是模具和设计变更成本高，维修和拆解复用较差。从 sustainable design 看，A 更符合 Reuse/Repair，B 更符合 Reduce/Dematerialization。4Rs 中 A 主要体现 Reuse 与 Repair，B 主要体现 Reduce。

原题 Q2(b): 20 个 current samples，要求 2 * Sigma control chart。

答案解析: 平均值 CL=502.85 mA，样本标准差 sigma=6.05 mA，2-sigma UCL=514.94 mA，LCL=490.76 mA。Sample 2 = 516 高于 UCL，Sample 17 = 490 低于 LCL，因此 process is not under control。



原题 Q2(c): target 500 mA, tolerance +/-5 mA，计算 Cp 与 Cpk。

答案解析: USL=505, LSL=495, mean=502.85, sigma=6.05。Cp=(USL-LSL)/(6sigma)=10/(36.28)=0.28；Cpk=min[(505-502.85)/(3sigma),(502.85-495)/(3sigma)]=0.12。Cp<1 且 Cpk 很低，说明 variation 过大且均值偏高，process capability 不足。

Q3 Project Management, Taguchi, CPM

原题 Q3(a)(i): O=8, M=10, P=13 months，三点估算。

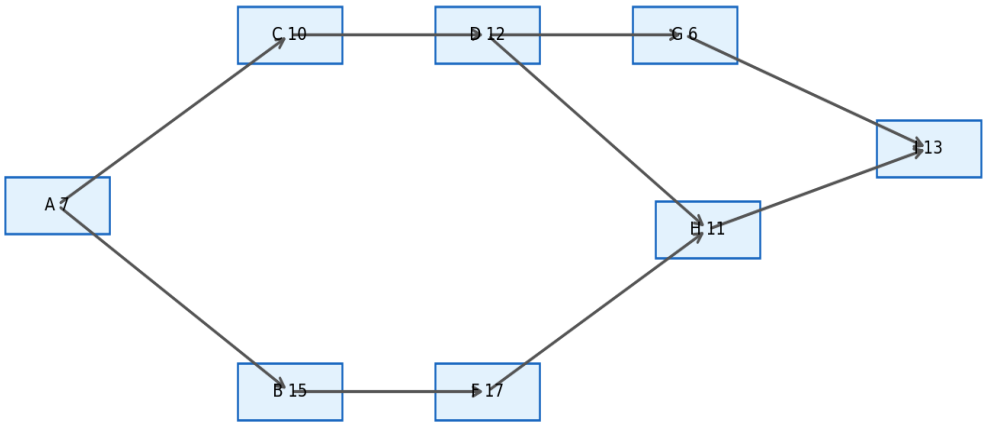
答案解析: Triangular expected duration=(8+10+13)/3=10.33 months，triangular sigma=sqrt((O^2+M^2+P^2-OM-OP-MP)/18)=1.03 months。Beta/PERT expected duration=(8+4*10+13)/6=10.17 months，PERT sigma=(13-8)/6=0.83 months。

原题 Q3(a)(ii)-(iv): communication channels, Taguchi loss, WBS。

答案解析: 按笔记口径把 project manager 也计入沟通人数，原人数=12，渠道=12*11/2=66；变化后 team members=12，加 PM 后=13，渠道=13*12/2=78，新增 12。Taguchi nominal-the-best, m=8.5, k=1，各 loss 为 0.1600, 0.1600, 0.0025, 0.5625, 0.1296, 0.0225, 0.0625, 0.0324, 0.1600, 0.3025；total loss=1.5945，average loss=0.15945。WBS 可写为 Level 0 Embedded temperature-sensor control system；Level 1 Planning, Design, Testing；Level 2 分别为 requirements/schedule-cost、hardware interface/software logic、unit test/integration test。

原题 Q3(b): CPM network A-I。

Set 1 Q3(b) CPM network



Act	Dur	ES	EF	LS	LF	Slack
A	7	0	7	0	7	0
B	15	7	22	7	22	0
C	10	7	17	17	27	10
D	12	17	29	27	39	10
F	17	22	39	22	39	0
G	6	29	35	44	50	15
H	11	39	50	39	50	0
I	13	50	63	50	63	0

答案解析: Critical Path = A -> B -> F -> H -> I，project duration = 63 days。

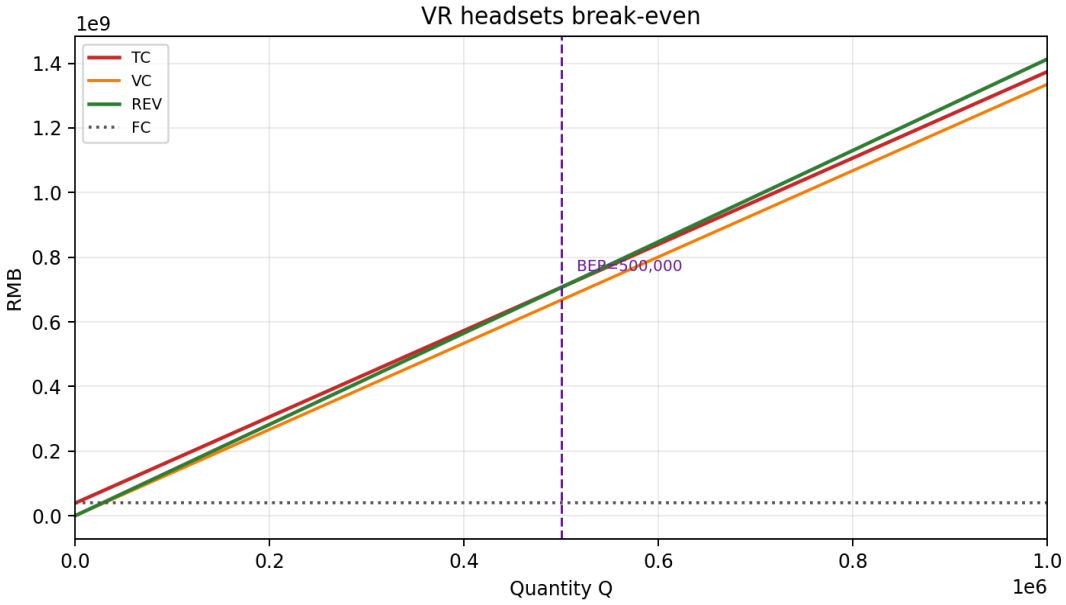
Q4 VRtuoso Make-or-Buy and Break-even

答案解析: 每年 0.6M headsets 需要 1.2M lenses。外购 supplier cost=675*1.2M=810M RMB。若设施出租，年度等效成本=810+39-2=847M，约 705.83 RMB/lens；若租约不续，=810+39=849M，约 707.50 RMB/lens。自制时 unit variable lens cost=325+215+127.5=667.5 RMB，annual variable=801M，加 fixed overhead 39M，总成本 840M，约 700 RMB/lens。因此在租约不续时应 manufacture internally，约省 9M RMB/year。

Headsets Q	VC	FC	TC	Revenue	Profit
0	0	39.0M	39.0M	0	-39.0M
0.2M	267.0M	39.0M	306.0M	282.6M	-23.4M

Headsets Q	VC	FC	TC	Revenue	Profit
0.4M	534.0M	39.0M	573.0M	565.2M	-7.8M
0.6M	801.0M	39.0M	840.0M	847.8M	7.8M
0.8M	1068.0M	39.0M	1107.0M	1130.4M	23.4M
1.0M	1335.0M	39.0M	1374.0M	1413.0M	39.0M

Full capacity profit=847.8M-840M=7.8M RMB。Break-even Q=39M/(1413-1335)=500,000 headsets。Safety margin=100,000 headsets=16.67%。



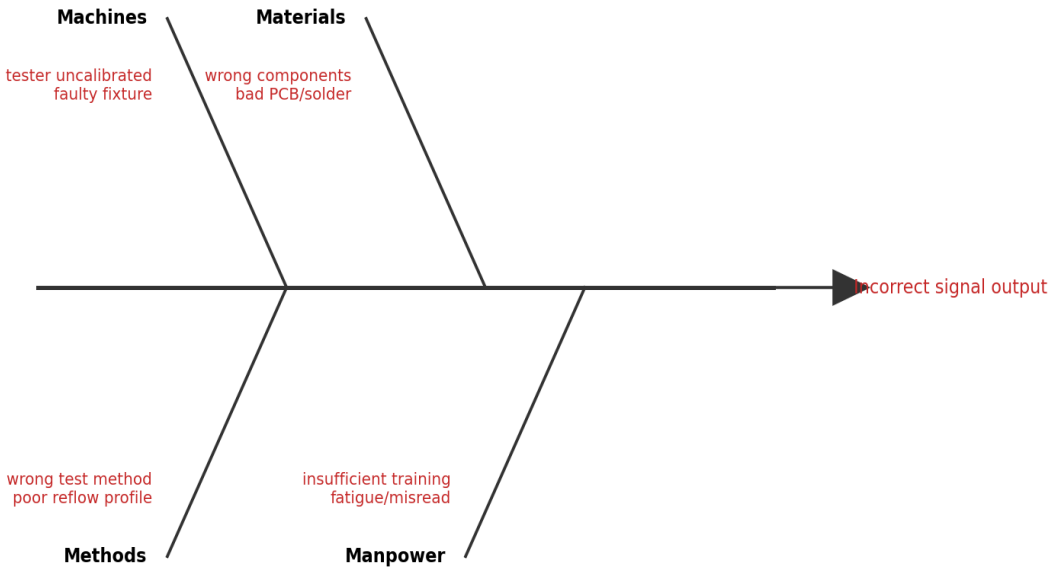
Sample Set 2

Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	DFM principles	C。减少零件数、使用简单形状可降低生产难度和成本。
1.2	Cause-effect diagram	D。Fishbone/Ishikawa 用图形组织问题的可能原因。
1.3	Process changes over time	I。Control Charts 用于观察过程随时间变化。
1.4	Not a project	H。周期性软件更新属于 operation，而不是 temporary unique project。
1.5	Risk technique	B。SWOT 用于识别内外部风险因素。
1.6	Official confirmation	C。Validate Scope 是对 deliverables 的正式验收。
1.7	Organizational evolution labels	C。1 Start-up, 2 Growth, 3 Expansion, 4 Formal Organisation。
1.8	Not resource efficient	C。把 waste 运到低环保标准地区不符合 sustainability。
1.9	Start-up structure true pair	D。Few layers 与 wide span of control 正确。
1.10	Hand tools vs automation	D。交叉点=14000/(1.0*5000)=2.8 years。
1.11	Biggest impact on price/quality/time	B。Product development 阶段决定 70-80% 影响。
1.12	New execution team meeting	B。优先复核 WBS 与职责。

Q2 Sustainability, Fishbone, BCR, Conflict

Q2(d): 三大 sustainability pillars 是 economic viability、environmental protection、social well-being。Manufacturing companies/designers 责任更大，因为设计阶段决定材料、能耗、维修性和回收性；end-user consumers 负责合理使用、维护、repair/reuse/recycle，但不能弥补不良设计。



Q2(e): Fishbone diagram 可按 Machines, Materials, Methods, Manpower 展开。Machines: tester uncalibrated, faulty fixture；Materials: wrong component values, bad PCB/solder；Methods: incorrect test procedure, poor reflow/profile；Manpower: insufficient training, fatigue or misreading。

Q2(f): BCR (Benefit-Cost Ratio)=total benefits/total costs。BCR=1 表示收益等于成本，财务上 break-even，除非有重要 intangible benefits，否则吸引力有限。WBS 是 Work Breakdown Structure。风险已分类后，下一步应评估 probability 和 impact，即 qualitative risk analysis。

Q2(g): (i) Withdraw/Avoid, lose-lose 或问题未解决；(ii) Smooth/Accommodate, PM lose/stakeholder win；(iii) Collaborate/Problem Solve, win-win；(iv) Force/Competing, PM win/engineer lose；(v) Compromise/Reconcile, both partial win and partial lose。

Q3 Discounted Cash Flow and Cash Flow

Q3(f): 三年费用为 1,000,000 + 1,050,000 + 1,102,500 = 3,152,500 RMB。若希望获得 today equivalent profit 100,000 RMB，交付时的 future value=100,000*(1.05)^3=115,762.5 RMB，所以 quote=3,268,262.5 RMB。

Year	Purchase	Maintenance	Power	Total actual cost
0	10.00M	-	0	10.00M
1	-	1.00M	0.50M	1.50M
2	-	1.00M	0.55M	1.55M
3	-	1.00M	0.605M	1.605M
4	-	1.00M	0.666M	1.666M
5	-	1.00M	0.732M	1.732M
Total	10.00M	5.00M	3.053M	18.053M

TCO: 按课堂源表口径，purchase 10M + PV maintenance 3.8M + PV power 2.5M = 16.3M RMB。严格逐年折现则约 16.06M RMB。

Q3(h): 累计收入 Y1-Y4=6025M，剩余 373M；Y5 每月 1480/12=123.33M，payback=4 years + 3.02 months，约 51 months。NPV=-6398 + 1400/1.05 + 1450/1.05^2 + 1550/1.05^3 + 1625/1.05^4 + 1480/1.05^5 = +85.98M RMB。NPV positive 且 5 年内回收，项目可考虑，但 margin 不大。

Q3(i): 30-day terms 下 cash-in 为上月 shipments，cash-out 为上月 components order。累计现金 M0=0, M1=-5, M2=-1, M3=7, M4=5, M5=-25 RMB。

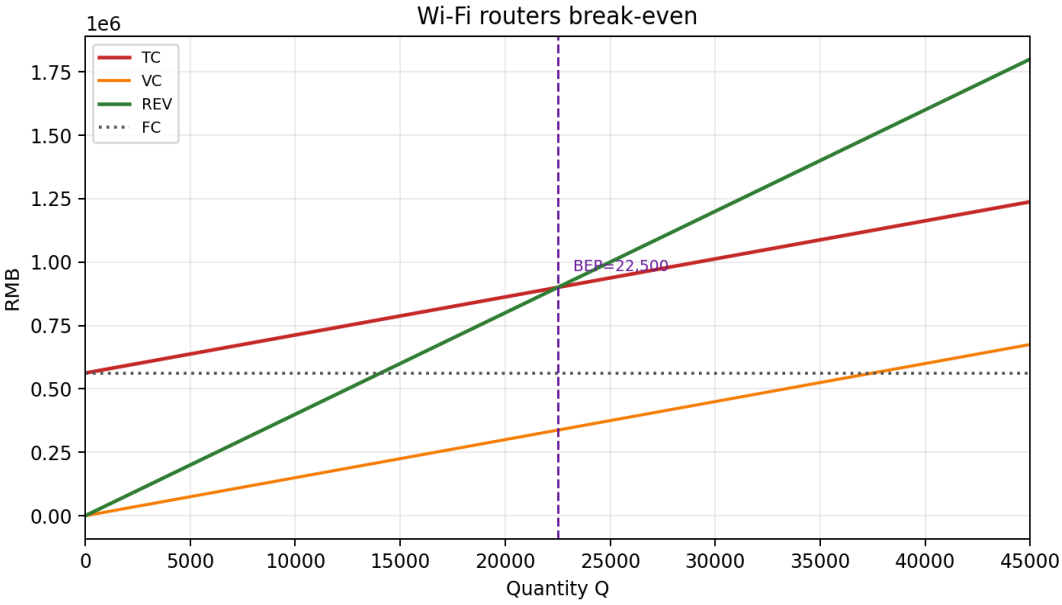
Q4 Wi-Fi Routers

Router unit VC = Direct Materials 9 + Direct Labour 5 + Variable Overhead 1 = 15 RMB/router；Fixed Overhead per full-capacity router=562,500/45,000=12.5 RMB。Make full cost=27.5 RMB/router。若购买且 fixed overhead 不可避免，buy equivalent cost=27.5+12.5=40 RMB/router，因此不建议外购。

若 facilities 可出租 570,000 RMB/year，则 buy annual equivalent=27.5*45,000 + 562,500 - 570,000 = 1,230,000 RMB，略低于 make 1,237,500 RMB；此时建议外购并出租设施，节省 7,500 RMB/year。

Q	VC	FC	TC	Revenue	Profit
10,000	150,000	562,500	712,500	400,000	-312,500
15,000	225,000	562,500	787,500	600,000	-187,500
20,000	300,000	562,500	862,500	800,000	-62,500
25,000	375,000	562,500	937,500	1,000,000	62,500
30,000	450,000	562,500	1,012,500	1,200,000	187,500

At full capacity Q=45,000, profit=40*45,000-(562,500+15*45,000)=562,500 RMB。Break-even Q=562,500/(40-15)=22,500 routers。Safety margin=45,000-22,500=22,500 routers=50%。


Wi-Fi routers break-even
1e6
TC
VC
REV
FC
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
RMB
BEP=22,500
0
5000
10000
15000
20000
25000
30000
35000
40000
45000
Quantity Q

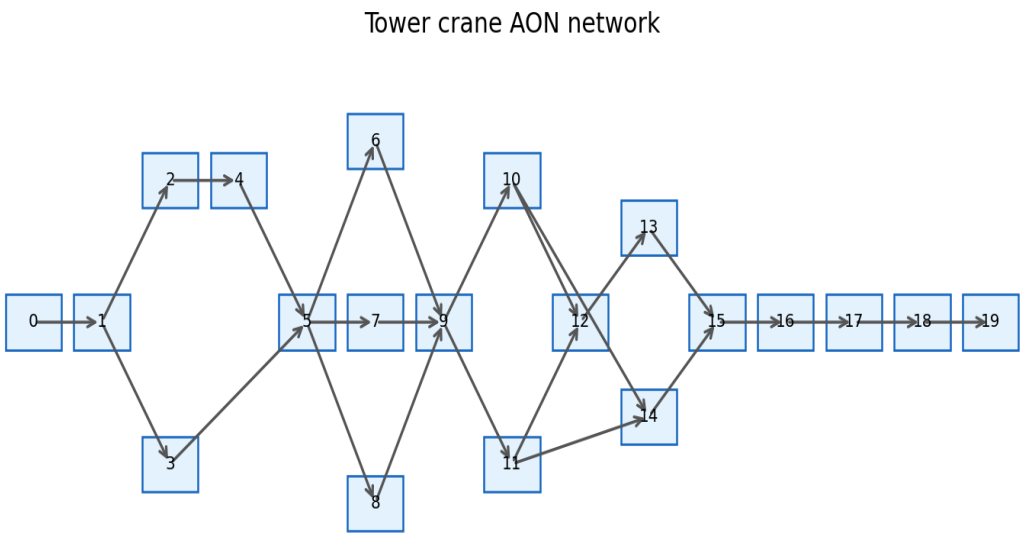
Sample Set 3

Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	Why project management	E。1,2,3,4 都是项目管理用途；5 只跟踪一次 schedule 错。
1.2	EBITDA	D。Gross Profit - operating expenses。
1.3	Organisation figure	B。Matrix Structure。
1.4	Sole Trader	A。个人经营，保留税后利润，也个人承担亏损。
1.5	Lease-or-buy	C。2000d = 30000+500d，d=20 days。
1.6	Break-even definition	B。Break-even at TC = R。
1.7	P&L statement	D。总结年度 revenues, payments and expenses。
1.8	Gross Profit	C。Sales - Cost of Sales。
1.9	Defender posture	B。Stability/efficiency, mechanistic, cost-efficient core technology。
1.10	DFM timing	C。Initial design has highest impact and lowest cost of change。
1.11	Sustainable principles except	D。Shorter usage is not sustainable。
1.12	Pareto chart	A。Descending bar chart by importance/frequency。

Q2 Tower Crane Project

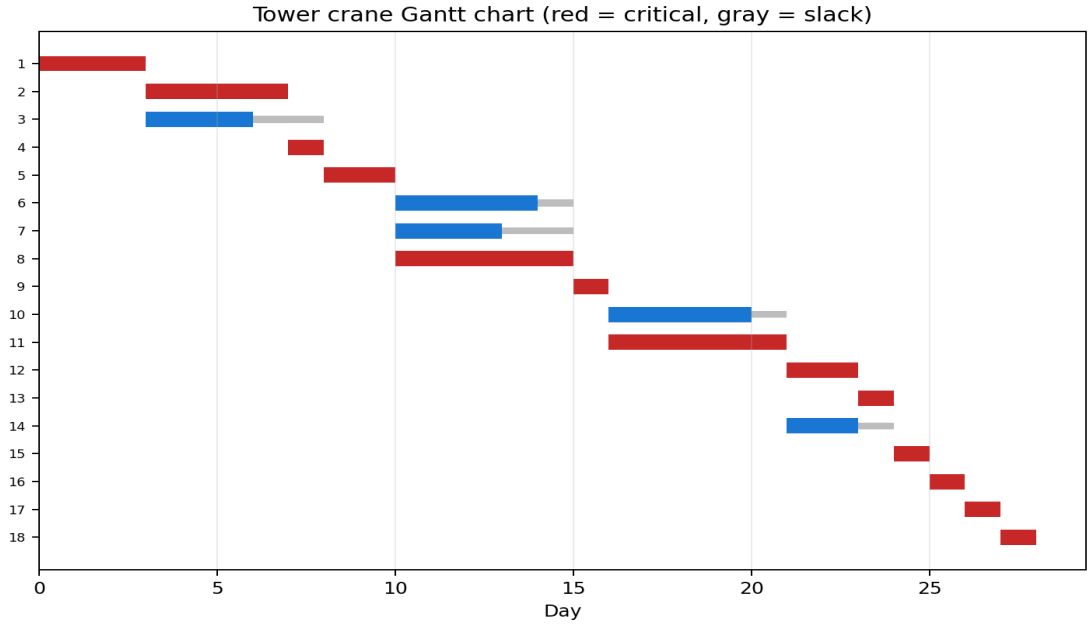
Network diagram 见下图。关键计算按 Activity-on-Node (AON) 做 forward pass/backward pass。



Task	ES	EF	LS	LF	Slack
1	0	3	0	3	0
2	3	7	3	7	0
3	3	6	5	8	2
4	7	8	7	8	0
5	8	10	8	10	0
6	10	14	11	15	1
7	10	13	12	15	2
8	10	15	10	15	0
9	15	16	15	16	0

Task	ES	EF	LS	LF	Slack
10	16	20	17	21	1
11	16	21	16	21	0
12	21	23	21	23	0
13	23	24	23	24	0
14	21	23	22	24	1
15	24	25	24	25	0
16	25	26	25	26	0
17	26	27	26	27	0
18	27	28	27	28	0

Critical Path = 0 -> 1 -> 2 -> 4 -> 5 -> 8 -> 9 -> 11 -> 12 -> 13 -> 15 -> 16 -> 17 -> 18 -> 19。Overall duration = 28 days。



若 Task 8 需等 Task 7 完成，则 Task 8 ES 从 day 10 改为 day 13，Task 9 之后整体推迟 3 天；new duration = 31 days。New critical path = 0 -> 1 -> 2 -> 4 -> 5 -> 7 -> 8 -> 9 -> 11 -> 12 -> 13 -> 15 -> 16 -> 17 -> 18 -> 19。Gantt chart 对项目经理的作用是可视化依赖、slack 和关键路径，变更发生时能快速看到哪些任务会推迟总工期。

Q3 DFM, QC, Robust Design, Six Sigma

Q3(h): DFM principles 可选 Simplicity: 减少零件和复杂形状；Standardization: 使用标准件和标准工艺；Tolerance: 在功能和成本间平衡公差；Material Selection: 兼顾成本、性能与可制造性。4Rs 可答 Reduce, Reuse, Repair, Recycle 任意两个。

Q3(i): 7-QC tools 可列 Flowchart, Check Sheet, Cause-Effect Diagram, Pareto Chart, Control Chart, Histogram, Scatter Diagram。监控 defect rate over time 用 Control Chart。若 process unstable，会看到点超出 UCL/LCL、连续趋势、runs 或非随机模式。

Q3(j): Taguchi loss functions 可答 Nominal-the-best, Smaller-the-better, Larger-the-better。Noise factors 可答 external/environmental noise 如温度湿度；unit-to-unit/manufacturing variation 如元件公差；deterioration/aging noise 如磨损老化。

Q3(k): DMAIC 五阶段为 Define, Measure, Analyze, Improve, Control。Six Sigma roles 可答 Executive Leadership 提供资源和方向；Champions 选择项目并移除障碍；Master Black Belts 培训和辅导；Black Belts 全职领导 DMAIC 项目；Green Belts 兼职支持改善。

Q4 Optical Transceivers

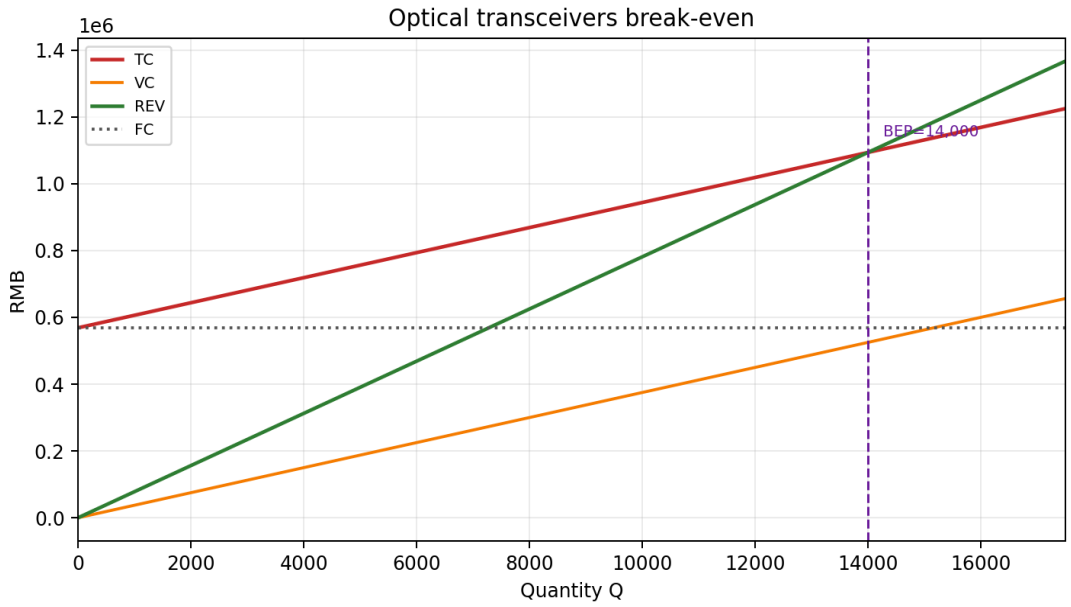
Break-even revenue relationship: TC=FC+VC*Q, REV=SP*Q。At break-even, FC+VC*Q=SP*Q，所以 Q=FC/(SP-VC)，Break-even Revenue=FC/[1-(VC/SP)]。

需求 17,500 transceivers。外购单价 57 RMB。设施出租时 equivalent annual cost=57*17,500 - 225,000 = 772,500 RMB，44.14 RMB/unit；租约到期后外购=997,500 RMB，57 RMB/unit。

自制 unit VC=22.5+12.5+2.5=37.5 RMB；annual VC=656,250；加 fixed overhead 568,750，总成本=1,225,000 RMB，70 RMB/unit。相比租约到期后外购 997,500 RMB，自制更贵，因此应继续购买。

Q	VC	FC	TC	Revenue	Profit
3,500	131,250	568,750	700,000	273,437.5	-426,562.5
7,000	262,500	568,750	831,250	546,875	-284,375
10,500	393,750	568,750	962,500	820,312.5	-142,187.5
14,000	525,000	568,750	1,093,750	1,093,750	0
17,500	656,250	568,750	1,225,000	1,367,187.5	142,187.5

Full capacity profit=142,187.5 RMB。Break-even Q=568,750/(78.125-37.5)=14,000 units。Safety margin=3,500 units=20%。



Sample Set 4

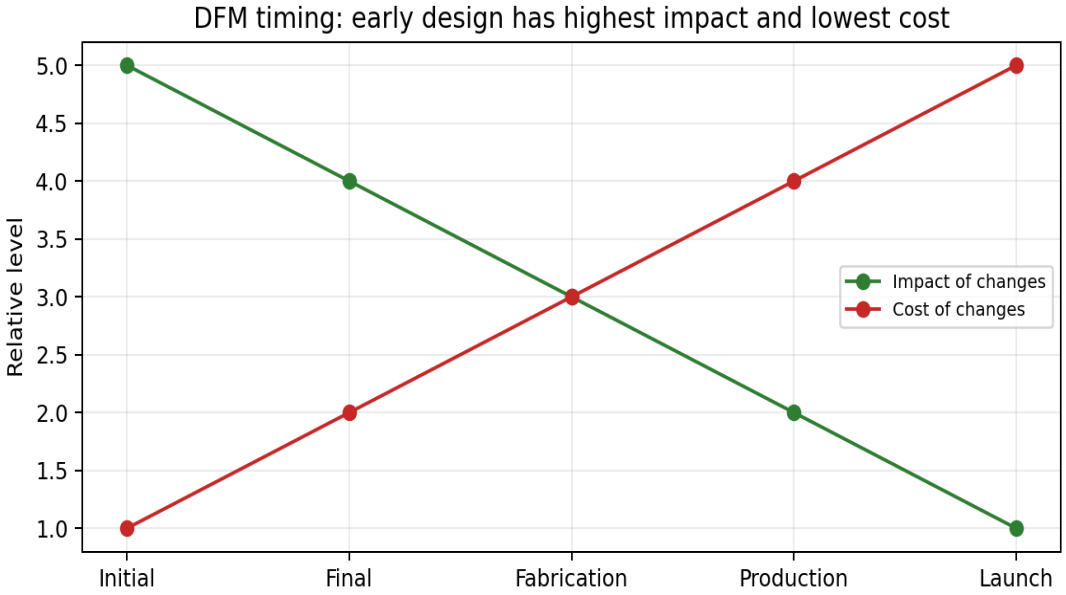
Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	DFM correct	G。70-80% production costs are determined by design decisions。
1.2	Cause-effect diagram	I。逻辑组织具体问题的可能原因。
1.3	Three pillars	L。Economic, environmental, social 三者平衡且不损害未来世代。
1.4	Not a project	K。Regular software updates 是 operation。
1.5	Initial phase not true	F。项目初期成本通常最低，不是最高。
1.6	WBS most correct	E。WBS includes 100% of work；not in WBS is out of scope。
1.7	Start-up true pair	D。Few layers + wide span of control。
1.8	Organisation labels	C。Start-up, Growth, Expansion, Formal Organisation。
1.9	Automation crossover	D。2.8 years。
1.10	Company valuation	A。300k-150k-100k+200k=250k CNY。
1.11	Risk management	D。Minimise threats and maximise opportunities。
1.12	Not risk response	E。Identification 属于 Identify Risks，不是 response strategy。

Q2 Discounted Cash Flow

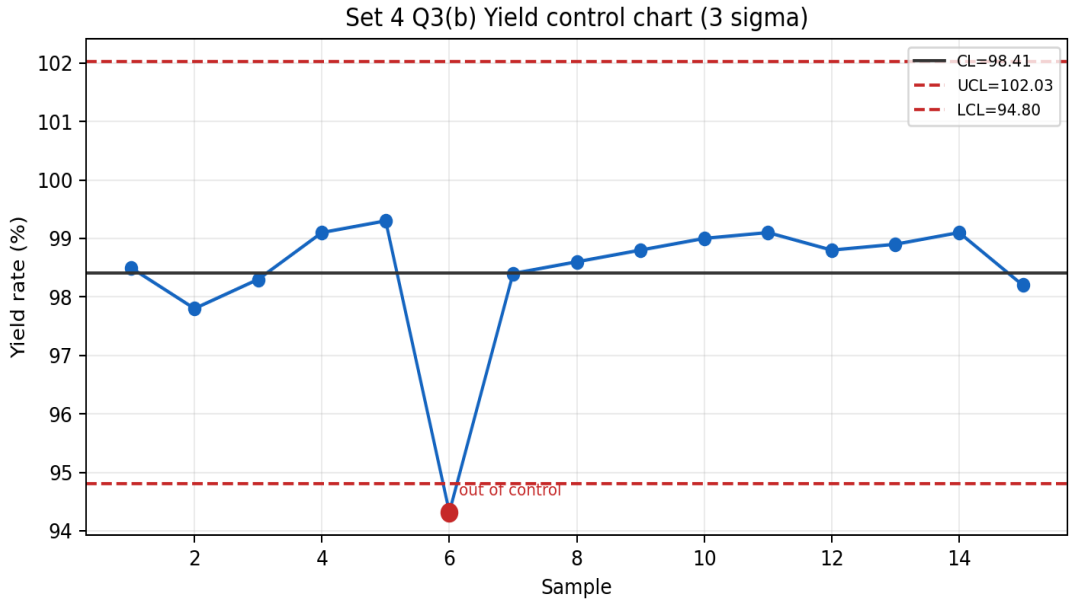
本题与 Set 2 Q3(f)-(i) 数值相同：quote=3,268,262.5 RMB；machine actual TCO=18.053M RMB，PV TCO 按课堂口径约 16.3M RMB；Chengdu facility payback 约 4 years 3 months，NPV=+85.98M RMB；现金流累计为 0, -5, -1, 7, 5, -25 RMB。

Q3 DFM, QC, Robust Design, Cpk



Q3(l): DFM 在 initial design 阶段最有效，因为 impact 最高且 cost of change 最低；越接近 launch，修改成本越高、影响越低。Sustainability pillars 可答 economic、environmental、social 任意两个。

Q3(m): Pareto Chart 是按 frequency/impact 降序排列的 bar chart；Control Chart 用于研究 process over time。Yield control chart 用 3-sigma: CL=98.415%, UCL=102.028%, LCL=94.801%；Sample 6=94.32% 低于 LCL，process unstable。

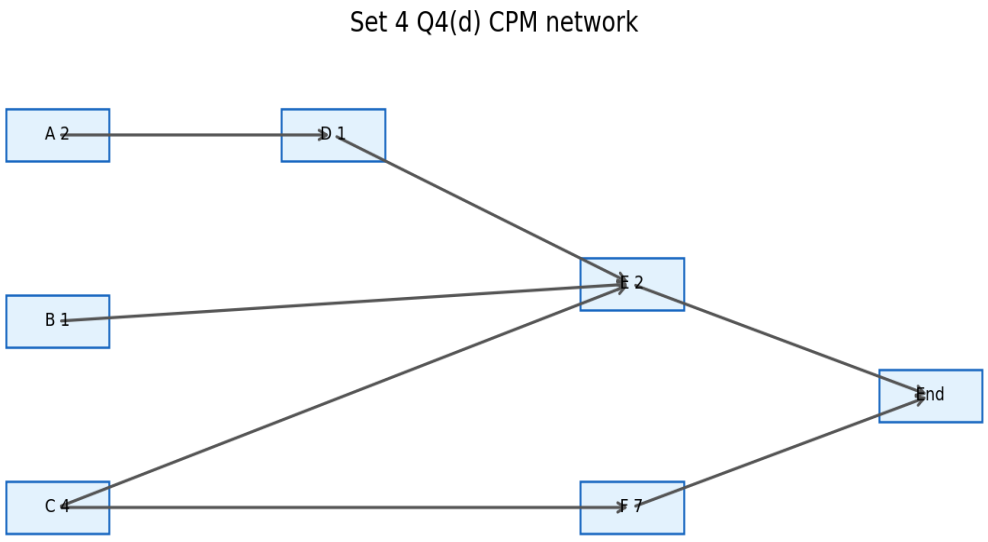


Q3(n): 题目说超出 110+/-20V 时半数情况需要 100 dollars repair，因此 functional-limit average loss A0=50。k=A0/Delta0^2=50/20^2=0.125；y=100 时 L=0.125*(100-110)^2=12.5 dollars。若课堂简化忽略 half，则会得 25 dollars，但本题文字更支持 12.5。Robust design 通过选择 control factors，使 output 对 noise factors 不敏感，从而减少 variation 并贴近 target。

Q3(o): USL=21, LSL=19, mean=19.8, sigma=0.2。Cpk=min[(21-19.8)/(3*0.2),(19.8-19)/(3*0.2)]=min(2.00,1.33)=1.33。Cp=2/(1.2)=1.67。Cpk 在 1.33 附近，低风险/可接受但均值偏向 LSL，应继续居中改善。

Q4 Project Management and CPM

Q4(c): BCR=1.4 表示 benefits 是 costs 的 1.4 倍，收益高于成本 40%，项目 financially attractive。Standard 是技术标准或最佳实践，通常可自愿采用；Regulation 是法律/监管要求，mandatory，违反会有法律后果。Basic Sender/Receiver model 主要含 sender, message, receiver, channel/noise, encode/decode；Interactive model 加入 feedback/response 和确认，是 two-way communication。



Act	Dur	ES	EF	LS	LF	Slack
A	2	0	2	6	8	6
B	1	0	1	8	9	8
C	4	0	4	0	4	0

Act	Dur	ES	EF	LS	LF	Slack
D	1	2	3	8	9	6
E	2	4	6	9	11	5
F	7	4	11	4	11	0

Small CPM: project duration=11 days，critical path=C -> F。若 C 延迟 1 天，因为 C slack=0，F 和项目完工也延迟 1 天，总工期变为 12 days；E 也受 C 影响但有 5 days float，不控制总工期。

Sample Set 5

Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	Critical path	A。Longest path with least/zero slack。
1.2	Start-up true pair	D。2 and 4。
1.3	Organisation labels	C。Start-up, Growth, Expansion, Formal Organisation。
1.4	More than one critical path	A。Risk increases because more paths have zero slack。
1.5	Scope decomposition document	C。Work Breakdown Structure (WBS)。
1.6	New team meeting	B。Review WBS and responsibilities。
1.7	Automation crossover	D。2.8 years。
1.8	DFM not correct	D。DFM 不应提高制造成本；其目标是 lower manufacturing costs。
1.9	Price/quality/manufacturing time impact	B。Product development 70-80%，manufacturing 20-30%。
1.10	Control chart definition	A。Determine whether process is stable/predictable over time。
1.11	Process capability	B。Consistently produce output meeting specifications。
1.12	Planning variables	A。Cost, Time, Performance。

Q2 Discounted Cash Flow

答案同 Set 2/4 的 DCF 题：quote=3,268,262.5 RMB；machine actual lifetime cost=18.053M RMB，PV TCO 约 16.3M RMB；facility payback 约 51 months，NPV=+85.98M RMB，attractive but sensitivity high。

Q3 DFM, QC, Robust Design, Six Sigma

Q3(p): DFM 最有效阶段是 initial design: highest impact, lowest cost of changes。Sustainability pillars 可列 economic/environmental/social 任意两个。

Q3(q): 7-QC tools 任意两个，如 Cause-Effect Diagram 与 Control Chart。Cause-effect diagram 的用法是把 effect 定义为 product returns due to defects，再按 Man/Machine/Material/Method/Measurement/Environment 分类头脑风暴 root causes，追到可行动原因。Training cost 属于 Prevention Cost；product recall and warranty repairs 属于 External Failure Cost。

Q3(r): Robust Design 三步可答 System Design, Parameter Design, Tolerance Design 任意两个。P Diagram 中 M=Signal Factor, X=Control Factor, Z=Noise Factor。Taguchi losses 与 Set 1 相同：total loss=1.5945，average loss=0.15945。

Q3(s): CTQ (Critical-to-Quality) 是显著影响输出质量、把 customer requirements 转为 measurable characteristics 的关键质量要素。Cpk for 20+/-1 mm, mean=19.8, sigma=0.2 is 1.33。

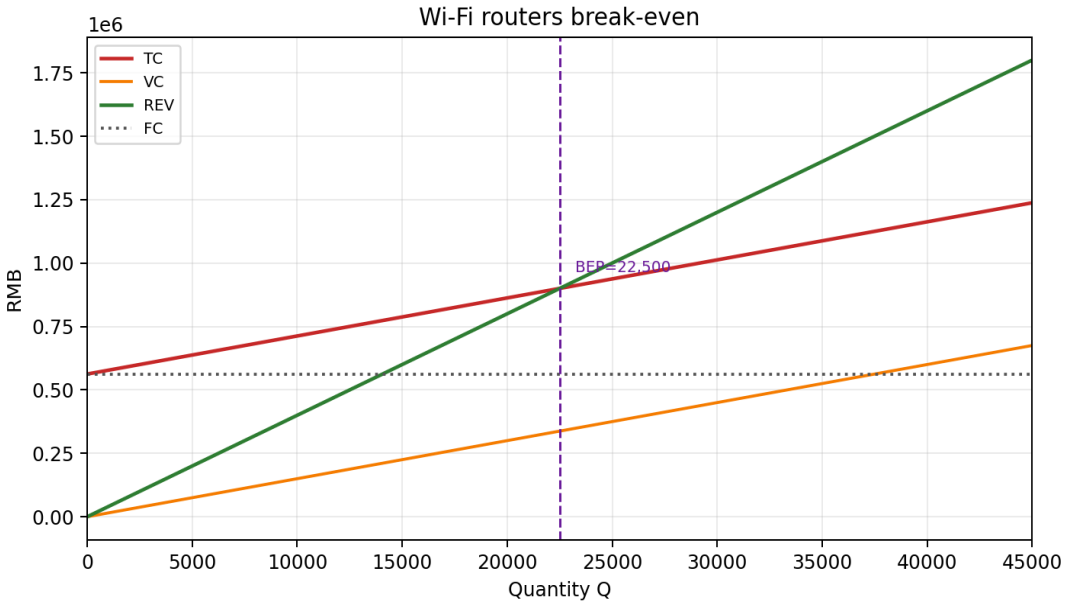
Q4 Wi-Fi Routers

Router unit VC = Direct Materials 9 + Direct Labour 5 + Variable Overhead 1 = 15 RMB/router；Fixed Overhead per full-capacity router=562,500/45,000=12.5 RMB。Make full cost=27.5 RMB/router。若购买且 fixed overhead 不可避免，buy equivalent cost=27.5+12.5=40 RMB/router，因此不建议外购。

若 facilities 可出租 570,000 RMB/year，则 buy annual equivalent=27.5*45,000 + 562,500 - 570,000 = 1,230,000 RMB，略低于 make 1,237,500 RMB；此时建议外购并出租设施，节省 7,500 RMB/year。

Q	VC	FC	TC	Revenue	Profit
10,000	150,000	562,500	712,500	400,000	-312,500
15,000	225,000	562,500	787,500	600,000	-187,500
20,000	300,000	562,500	862,500	800,000	-62,500
25,000	375,000	562,500	937,500	1,000,000	62,500
30,000	450,000	562,500	1,012,500	1,200,000	187,500

At full capacity Q=45,000, profit=40*45,000-(562,500+15*45,000)=562,500 RMB。Break-even Q=562,500/(40-15)=22,500 routers。Safety margin=45,000-22,500=22,500 routers=50%。



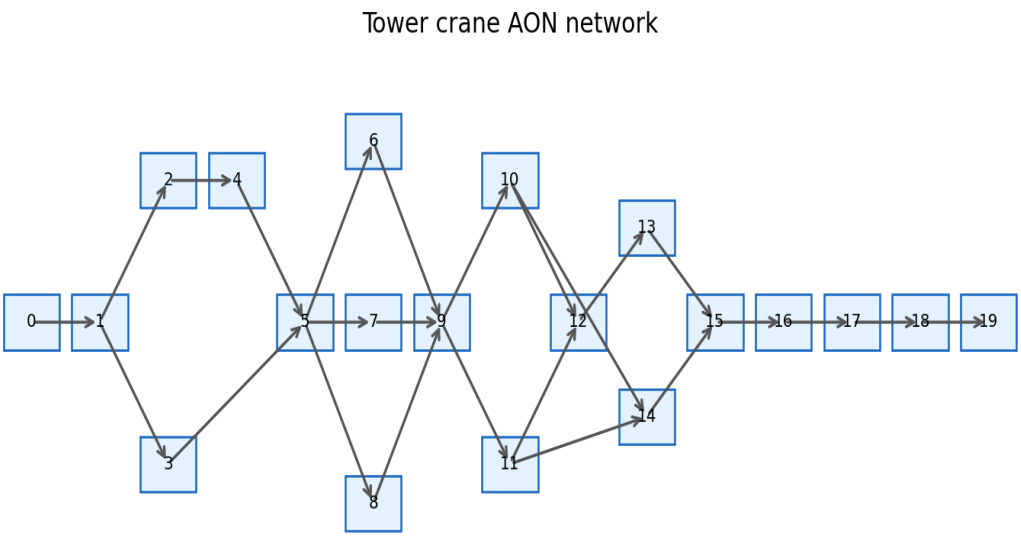
Sample Set 6

Q1 Multiple Choice

题号	原题关键词	答案与中文解析
1.1	DFM initial vs production	F。Initial design has higher impact and lower change cost。
1.2	Sustainable principles except	I。Design for shorter usage is not sustainable。
1.3	Not appraisal cost	D。Equipment upgrades 不属于 appraisal cost。
1.4	NPV/BCR/IRR attractiveness	J。异质指标不能直接判定最优。
1.5	SWOT	G。Risk identification tool/technique。
1.6	CPI=1.1, SPI=0.9	G。Under budget but behind schedule。
1.7	EBITDA	D。Gross Profit - operating expenses。
1.8	Organisation figure	B。Matrix Structure。
1.9	Sole Trader	A。Individual business with personal responsibility for losses。
1.10	Lease-or-buy	C。20 days。
1.11	P&L statement	D。Summarises revenues, payments and expenses。
1.12	Defender posture	B。Stability/efficiency, mechanistic, one core cost-efficient technology。

Q2 Tower Crane, Process Groups, Leadership, EVM

Network diagram 见下图。关键计算按 Activity-on-Node (AON) 做 forward pass/backward pass。



Task	ES	EF	LS	LF	Slack
1	0	3	0	3	0
2	3	7	3	7	0
3	3	6	5	8	2
4	7	8	7	8	0
5	8	10	8	10	0
6	10	14	11	15	1
7	10	13	12	15	2
8	10	15	10	15	0
9	15	16	15	16	0

Task	ES	EF	LS	LF	Slack
10	16	20	17	21	1
11	16	21	16	21	0
12	21	23	21	23	0
13	23	24	23	24	0
14	21	23	22	24	1
15	24	25	24	25	0
16	25	26	25	26	0
17	26	27	26	27	0
18	27	28	27	28	0

Critical Path = 0 -> 1 -> 2 -> 4 -> 5 -> 8 -> 9 -> 11 -> 12 -> 13 -> 15 -> 16 -> 17 -> 18 -> 19。Overall duration = 28 days。

若 Task 8 需等 Task 7 完成，则 Task 8 ES 从 day 10 改为 day 13，Task 9 之后整体推迟 3 天；new duration = 31 days。New critical path = 0 -> 1 -> 2 -> 4 -> 5 -> 7 -> 8 -> 9 -> 11 -> 12 -> 13 -> 15 -> 16 -> 17 -> 18 -> 19。Gantt chart 对项目经理的作用是可视化依赖、slack 和关键路径，变更发生时能快速看到哪些任务会推迟总工期。

Q2(c): (i) Monitoring and Controlling process group；(ii) Planning process group；(iii) Initiating process group；(iv) Closing process group。

Q2(d): Leadership 关注 vision、motivation、change 和 people；Management 关注 planning、organizing、controlling 和 process。Leadership 倾向 inspire and align，Management 倾向 assign, monitor and correct。Leadership 更长期和方向性，Management 更短期和执行性。

Q2(e): SPI=EV/PV=8500/10000=0.85 <1，behind schedule。CPI=EV/AC=8500/8000=1.0625 >1，under budget/cost efficient。

Q3 DFM/Six Sigma/PM

Q3(t): Poka-yoke 是 mistake proofing，用设计防止错误发生或立刻暴露错误。例如 LED 实验夹具使用 keyed connector 和 polarity guide，只有 LED 极性、电阻和供电连接正确时 green indicator 才亮，反接则无法插入或 red warning。Resistor capability: USL=14, LSL=10, mean=12.2, sigma=0.5。Cp=4/(3)=1.33；Cpk=min(1.8/1.5,2.2/1.5)=1.20。潜在能力勉强，但均值偏高，actual capability 中等风险，需要 centering improvement。4Rs 可答 Reduce, Reuse, Repair, Recycle 任意两个。

Q3(u): Talent Triangle 可选 Technical Project Management、Strategic and Business Management、Leadership 任意两个并解释。WBS 是把 project/product scope 分解为可管理组件的层级结构。Sarah 三点估算: O=9, M=12, P=16。Triangular E=12.33 weeks, sigma≈1.43；Beta E=(9+4*12+16)/6=12.17 weeks, sigma=(16-9)/6=1.17。Communication channels 按笔记口径计入 PM: 原 PM+6=7 人，21 channels；新 PM+7=8 人，28 channels；新增 7。Conflict: win-win is Collaborate/Problem Solve；lose-lose is Compromise/Reconcile because both give up something；neither side gains nor loses is Withdraw/Avoid because issue is postponed/unresolved。

Q4 Optical Transceivers

Break-even revenue relationship: TC=FC+VC*Q, REV=SP*Q。At break-even, FC+VC*Q=SP*Q，所以 Q=FC/(SP-VC)，Break-even Revenue=FC/[1-(VC/SP)]。

需求 17,500 transceivers。外购单价 57 RMB。设施出租时 equivalent annual cost=57*17,500 - 225,000 = 772,500 RMB，44.14 RMB/unit；租约到期后外购=997,500 RMB，57 RMB/unit。

自制 unit VC=22.5+12.5+2.5=37.5 RMB；annual VC=656,250；加 fixed overhead 568,750，总成本=1,225,000 RMB，70 RMB/unit。相比租约到期后外购 997,500 RMB，自制更贵，因此应继续购买。

Q	VC	FC	TC	Revenue	Profit
3,500	131,250	568,750	700,000	273,437.5	-426,562.5
7,000	262,500	568,750	831,250	546,875	-284,375
10,500	393,750	568,750	962,500	820,312.5	-142,187.5
14,000	525,000	568,750	1,093,750	1,093,750	0

Q	VC	FC	TC	Revenue	Profit
17,500	656,250	568,750	1,225,000	1,367,187.5	142,187.5

Full capacity profit=142,187.5 RMB。Break-even Q=568,750/(78.125-37.5)=14,000 units。Safety margin=3,500 units=20%。

